Dissertation Projects for 2025-26
Mathematics Pathway
Data Science MSc[1]

[1] By default, each supervisor takes one student; higher capacities are shown in brackets next to their names.

M1: Assessing Gerrymandering

Proposer: Mark Muldoon (up to 3), M.Muldoon@Manchester.ac.uk

Background

Gerrymandering is the process of drawing electoral boundaries in such a way as to enhance the legislative power of one party at the expense of others: it has a long history in the United States. Although Federal law requires all the Congressional Districts[2] within a given state to have populations that are as nearly equal "as is practicable", the states still have considerable freedom in determining the *shapes* of districts. This has lead to some remarkably convoluted boundaries, such as those illustrated below.

The drawing of electoral districts has become a data science problem: the party in power when redistricting occurs, armed with detailled voting histories and demographic data, seeks to freeze-in a decade-long advantage. The precision with which such partisan gerrymandering can now be done has generated considerable controversy, including numerous law suits, some of which have reached the American Supreme Court. There are also many initiatives and articles aimed at the general reader, for example [Kla17; Mgg], as well as a substantial academic literature that includes work by lawyers; economists, [FH08]; mathematicians, [DT18; BD17] and statisticians, [Her+18; HRM17; Ban+17].

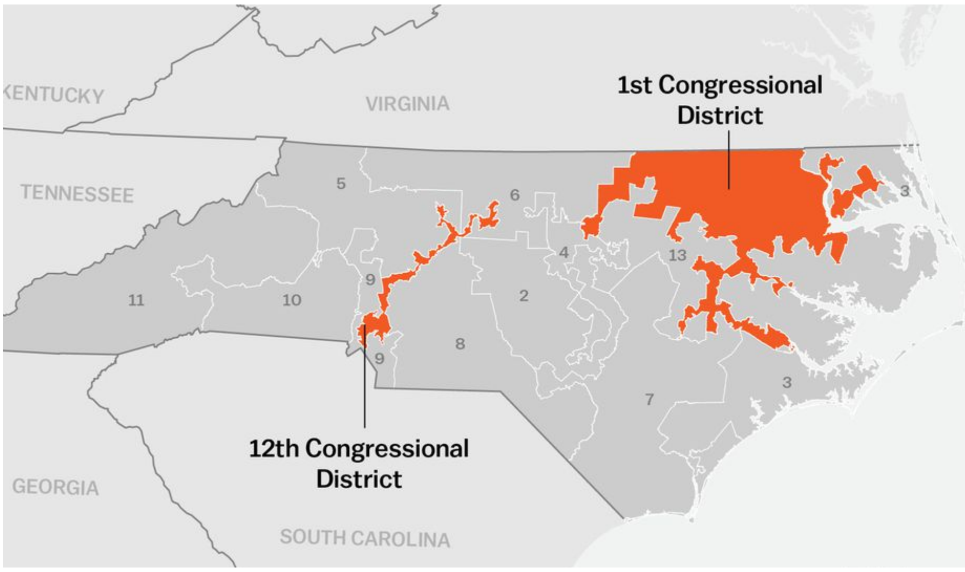

Figure 1: *This figure, which is adapted from one that appeared in the online magazine* Vox *[Ygl18], shows the boundaries of North Carolina's 1st and 12th congressional district as drawn after the 2010 census.*

Description

Mathematical work on gerrymandering has focussed on ways to detect and quantify it, with some authors considering the shape of the districts—asking, for example, whether they have smooth boundaries and are "compact" in the sense that most voters live near the center of the district—while others have taken a more statistical, hypothesis-testing, view and asked whether the existing district boundaries appear exceptional in some suitably-defined ensemble of all possible boundaries. In this project you'd seek to understand and replicate some of these approaches, then adapt them to some other political setting.

[2] The American national legislature has two components, the House of Representatives and the Senate. There are 435 members of the House and they are shared-out among the 50 states after each census, which happens once per decade.

Deliverables

- Assemble, clean and organise a data set that allows one to simulate an election with a given set of electoral district boundaries.
- Understand, apply and critique various measures of gerrymandering as applied to the assembled data.

Prerequisites and skills

Some interest in and knowledge of electoral politics is essential, as are solid programming skills. An interest in Markov Chain Monte Carlo (MCMC) methods[3] would be helpful.

References

[DT18] M Duchin and BE Tenner. *Discrete geometry for electoral geography*. 2018. arXiv: `1808.05860` (physics.soc-ph).

[Her+18] G Herschlag, HS Kang, J Luo, CV Graves, S Bangia, R Ravier, and JC Mattingly. *Quantifying Gerrymandering in North Carolina*. 2018. arXiv: `1801.03783` (stat.AP).

[Mgg] *Metric Geometry and Gerrymandering Group*. 2018. URL: `https://mggg.org` (Accessed 06 Dec. 2018).

[Ygl18] M Yglesias. "It's harder for Democrats to gerrymander effectively". In: *Vox* (2018). URL: `https://www.vox.com/policy-and-politics/2018/4/2/17173158/democrats-gerrymander-segregation`.

[Ban+17] S Bangia, C Vaughn Graves, G Herschlag, HS Kang, J Luo, JC Mattingly, and R Ravier. *Redistricting: Drawing the Line*. 2017. arXiv: `1704.03360` (stat.AP).

[BD17] M Bernstein and M Duchin. *A formula goes to court: Partisan gerrymandering and the efficiency gap*. 2017. arXiv: `1705.10812` (physics.soc-ph).

[HRM17] G Herschlag, R Ravier, and JC Mattingly. *Evaluating Partisan Gerrymandering in Wisconsin*. 2017. arXiv: `1709.01596` (stat.AP).

[Kla17] E Klarreich. "How to Quantify (and Fight) Gerrymandering". In: *Quanta Magazine* (2017). URL: `https://www.quantamagazine.org/the-mathematics-behind-gerrymandering-20170404/`.

[FH08] JN Friedman and RT Holden. "Optimal Gerrymandering: Sometimes Pack, But Never Crack". In: *American Economic Review* 98.1 (2008), pp. 113–144. DOI: `10.1257/aer.98.1.113`.

[3] An introduction to this topic forms part of Stats & Machine Learning II.

M2: Mixtures of Markov Chains

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Background

Suppose we have data in the form of S sequences, each of the form

$$x_{s,0}, x_{s,1}, \ldots, x_{s,t}, \ldots x_{s,n_s}$$

where

- $x_{s,t} \in \{1, \ldots, M\}$ is a state;
- the first subscript, $s \in \{1, \ldots, S\}$, indicates which of the S observed chains is under discussion;
- and the second subscript, $t \in \{0, \ldots, n_s\}$, is a time.

That is, $x_{s,t} =$ the state of the sequence s at time t.

Such sequences might arise from:

- self-assessed mood, pain and other measures, all recorded daily on a five-point scale. A recent PhD student worked on such data from the Cloudy with a Chance of Pain study [Das+22].
- logs recorded by web-based apps that capture the sequence of steps taken by users of the site: this is an example suggested in [GKV16].
- time-stamped GPS data used to characterise patterns of travel in and around Nanjing [Zho+21].

All these various sequences can be modelled with discrete-time, finite-state Markov processes. You can think of such a process as a random walk on a finite, directed graph: the states of the chain form the vertex set of the graph while edges—which have probabilities associated with them—reflect the allowed state-to-state transitions.

Description

In this project you would consider fitting *mixtures* of Markov Chains[4] as a way to, simultaneously, characterise and classify the underlying processes that generate the sequences. For example, in the case of the log files from the web-based app, you might want to determine whether different types of users use the app differently: experienced users might be more efficient, or have different goals, than beginners.

Deliverables

- Find a real-world data set suited to this class of models. In 2021–22 Lintao Wang looked at Transport For London's cycle hire data.
- Develop codes using Stan [Sta21] to fit such models.

Prerequisites and skills

It would be helpful if you already knew something about Markov Chains, though if you need to learn this material there are many excellent introductory books, including [Bré20]. Solid programming skills in R (ideally) or Python are necessary and an interest in Markov Chain Monte Carlo (MCMC) methods[5] would also be helpful.

[4]We will learn about mixture models—which are related to Kernel Density Estimates—during the lectures in Stats & Machine Learning II devoted to classification.

[5] An introduction to this topic forms part of Stats & Machine Learning II.

References

[Das+22] R Das, M Muldoon, M Lunt, J McBeth, BB Yimer, and T House. *Modelling and classifying joint trajectories of self-reported mood and pain in a large cohort study*. 2022.

[Sta21] Stan Development Team. *Stan Modeling Language Users Guide and Reference Manual, 2.28*. 2021. URL: https://mc-stan.org.

[Zho+21] Y Zhou, Q Yuan, C Yang, and Y Wang. "Who you are determines how you travel: Clustering human activity patterns with a Markov-chain-based mixture model". In: *Travel Behaviour and Society* 24 (2021), pp. 102–112. ISSN: 2214-367X. DOI: 10.1016/j.tbs.2021.03.005.

[Bré20] P Brémaud. *Markov Chains: Gibbs Fields, Monte Carlo Simulation, and Queues*. 2nd edition. Vol. 31. Texts in Applied Mathematics. Springer, Cham, 2020. ISBN: 978-0-387-98509-1. DOI: 10.1007/978-3-030-45982-6.

[GKV16] R Gupta, R Kumar, and S Vassilvitskii. "On Mixtures of Markov Chains". In: *Advances in Neural Information Processing Systems 29 (NIPS 2016)*. Ed. by DD Lee, M Sugiyama, UV Luxburg, I Guyon, and R Garnett. Curran Associates, Inc., 2016, pp. 3441–3449. URL: http://papers.nips.cc/paper/6078-on-mixtures-of-markov-chains.pdf.

M3: Kolmogorov-Arnold representations for deep learning

Proposer: Mark Muldoon (up to 3), M.Muldoon@Manchester.ac.uk

Background

Hilbert's thirteenth problem asks whether the roots of the seventh-order polynomial equation

$$x^7 + ax^3 + bx^2 + cx + 1 = 0,$$

when considered as functions of the three coefficients a, b and c, can be represented as a composition of a finite number of continuous two-variable functions [Hil02]. There is also a variant of the problem that asks about compositions of algebraic functions. Kolmogorov subsequently proved [Kol57] that every continuous, real-valued function $f(x_1, \ldots, x_n)$ on the n-dimensional unit cube can be represented as a composition of one-dimensional, real-valued functions and addition. More concretely, he proved that for every continuous function $f : [0,1]^n \to \mathbb{R}$, there exist continuous real functions $\psi_{p,q} : [0,1] \to \mathbb{R}$ and continuous real functions $\Phi_q : \mathbb{R} \to \mathbb{R}$ such that

$$f(x_1, \ldots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \psi_{p,q}(x_p) \right). \tag{1}$$

Arnold later proved that every continuous real function of three variables, defined on a three-dimensional unit cube, can be represented in terms of continuous real functions of two variables [Arn09]. The theorems of Kolmogorov and Arnold combine to answer in the affirmative the continuous variant of Hilbert's thirteenth problem and their solution is now known as the Kolmogorov-Arnold (KA) theorem.

Numerous modifications and extensions of the KA theorem have since been developed including, for example, constructive proofs. Further, the Kolmogorov superposition and its variants have been interpreted as feed-forward neural networks [HN87; Köp02]. The constructive proofs and the neural network interpretations of the KA theorem provide scope for KA-based machine learning algorithms for function approximation, a topic that has recently attracted considerable attention in both the popular [Bis24; Nad24] and scholarly [Liu+24; LS24] literature.

Description

The functions $\psi_{p,q}$ of the KA representation in Eqn. (1) are known as *inner functions* and when designing a KA-network, one can require that they satisfy various conditions, such as continuity and smoothness. Alternatively, using ideas of [GM19] and [SD02], one can design the inverses $\psi_{p,q}^{-1}$ of the inner functions, which turn out to be space-filling curves.

Deliverables

- Understand the algorithms of [Spr65] and [Köp02].
- Reproduce results of [Liu+24].
- Understand the connection between KA-networks and space-filling curves, as developed in [GM19; SD02] and develop KA-networks based on novel space-filling curves.

Pre-requisites and skills

The only pre-requisite is knowledge of Python. If you want to go deeply into the properties of space-filling curves in the context of KA approximation, some knowledge of mathematical analysis and probability would be helpful.

References

[Bis24] M Bischoff. *An Alternative to Conventional Neural Networks Could Help Reveal What AI Is Doing behind the Scenes*. 2024. URL: https://www.scientificamerican.com/article/an-alternative-to-conventional-neural-networks-could-help-reveal-what-ai-is/ (Accessed 28 May 2024).

[LS24] M-J Lai and Z Shen. *The Kolmogorov Superposition Theorem can Break the Curse of Dimensionality When Approximating High Dimensional Functions*. 2024. arXiv: 2112.09963 (math.NA).

[Liu+24] Z Liu, Y Wang, S Vaidya, F Ruehle, J Halverson, M Soljačić, TY Hou, and M Tegmark. *KAN: Kolmogorov-Arnold Networks*. 2024. arXiv: 2404.19756 (cs.LG).

[Nad24] S Nadis. "Novel Architecture Makes Neural Networks More Understandable". In: *Quanta Magazine* (2024). URL: https://www.quantamagazine.org/novel-architecture-makes-neural-networks-more-understandable-20240911/.

[GM19] A Gorchakov and V Mozolenko. "Analysis of approaches to the universal approximation of a continuous function using Kolmogorov's superposition". In: *2019 International Conference on Engineering and Telecommunication (EnT)*. IEEE. 2019, pp. 1–4. DOI: 10.1109/EnT47717.2019.9030591.

[Arn09] VI Arnol'd. "On the representation of functions of several variables as a superposition of functions of a smaller number of variables". In: *Collected Works: Representations of Functions, Celestial Mechanics and KAM Theory, 1957–1965*. Ed. by AB Givental, BA Khesin, JE Marsden, AN Varchenko, VA Vassiliev, OY Viro, and VM Zakalyukin. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 25–46. ISBN: 978-3-642-01742-1. DOI: 10.1007/978-3-642-01742-1_5.

[Köp02] M Köppen. "On the Training of a Kolmogorov Network". In: *Artificial Neural Networks - ICANN 2002*. Ed. by JR Dorronsoro. Vol. 2415. Lecture Notes in Computer Science. Madrid, Spain: Springer-Verlag Heidelberg, 2002, pp. 474–479. DOI: 10.1007/3-540-46084-5_77.

[SD02] DA Sprecher and S Draghici. "Space-filling curves and Kolmogorov superposition-based neural networks". In: *Neural Networks* 15.1 (2002), pp. 57–67. ISSN: 0893-6080. DOI: https://doi.org/10.1016/S0893-6080(01)00107-1. URL: https://www.sciencedirect.com/science/article/pii/S0893608001001071.

[HN87] R Hecht-Nielsen. "Kolmogorov's mapping neural network existence theorem". In: *Proceedings of the international conference on Neural Networks*. Vol. 3. IEEE Press New York. 1987, pp. 11–14.

[Spr65] DA Sprecher. "On the structure of continuous functions of several variables". In: *Transactions of the American Mathematical Society* 115 (1965), pp. 340–355. DOI: 10.1090/S0002-9947-1965-0210852-X.

[Kol57] A Kolmogorov. "On the representation of continuous functions of several variables by superpositions of continuous functions of lesser variable count". In: *Doklady Akademii Nauk SSSR* 114.5 (1957), pp. 953–956.

M4: Bayesian inference for mosquito control tool evaluation models

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Background

Mosquito-borne diseases such as malaria and dengue pose significant public health challenges globally. Mathematical models of mosquito behaviour are essential for understanding transmission dynamics and evaluating vector control interventions.

A key challenge in fitting these models to trial data is the appropriate handling of parameter uncertainty and model structure. Bayesian inference methods, particularly Markov Chain Monte Carlo (MCMC), offer a rigorous framework for parameter estimation, but model parameterisation choices significantly affect the quality of inference. Issues such as overdispersion in count data from trap catches and random effects to account for variation can all impact model identifiability and predictive accuracy.

Description

This project will explore how different modelling choices affect parameter estimation and uncertainty quantification. Key questions could include:

- How does incorporating random effects improve model fit and parameter identifiability?
- What are drivers of overdispersion in trial data?
- Can hierarchical Bayesian models better capture heterogeneity?
- Could experiments be improved to reduce uncertainty and bias?

The student will implement these models using modern Bayesian inference tools (such as Stan) and compare approaches using simulated and/or real trial data.

Deliverables

- Implementation of baseline mosquito behaviour model with standard parameterisation
- Simulation study assessing parameter identifiability and uncertainty quantification under different model and data structures
- Application to real mosquito control product evaluation data

Prerequisites and skills:

- Strong programming skills (R or Python required; experience with Stan, PyMC, or C/C++ programming languages advantageous)
- Understanding of Bayesian inference and MCMC methods (covered in Stats & Machine Learning II)

References

[1] Fairbanks EL, Tambwe MM, Moore J, Mpelepele A, Lobo NF, Mashauri R, Chitnis N, Moore SJ. (2024) *Evaluating human landing catches as a measure of mosquito biting and the importance of considering additional modes of action*. Scientific Reports. 14(1):11476. https:/doi.org/10.1038/s41598-024-61116-0

[2] Vajda ÃĽA, Ross A, Doum D, Fairbanks EL, Chitnis N, Hii J, Moore SJ, Richardson JH, Macdonald M, Sovannaroth S, Kimheng P. (2024). *Field evaluation of a volatile pyrethroid spatial repellent and etofenprox treated clothing for outdoor protection against forest malaria vectors in Cambodia*. Scientific Reports. 14(1):17348. https:/doi.org/10.1038/s41598-024-67470-3

M5: Comparison of Bayesian regression model covariates for predicting asymptomatic Covid-19 infection from activity diary data

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Background

Identifying which settings and activities pose the greatest risk for infectious disease transmission is crucial for targeted public health interventions. During the COVID-19 pandemic, activity diary data combined with infection testing provided opportunities to quantify transmission risks across different settings (e.g., households, workplaces, social gatherings, public transport).

Previous work focused on the methodological question: what measure of exposure best predicts infection risk? Candidates included:

- Binary participation: Did someone participate in an activity? (yes/no)
- Contact count: Number of non-household contacts during the activity
- Duration: Time spent in the activity
- Contact-time product: Number of contacts ÃŮ duration

However, the optimal exposure metric may vary by setting. For example, brief high-contact events (crowded public transport) may have different risk profiles than prolonged low-contact activities (outdoor exercise). Furthermore, traditional analyses typically focus on non-household contacts, but the presence of household members during activities could modify risk.

Description

This project will use Bayesian regression models and model comparison techniques to analyse test data and activity diary data. The student will fit separate Bayesian regression models using each exposure metric (binary, contacts, duration, contact-time product) and compare models using formal model selection criteria

Students may then extend the model to answer questions such as:

- Does varying the exposure metric by setting improve predictions for infection risk?
- What is the impact of household contact co-participation in activities?

Deliverables

- Application to real data from UK university cohort
- Implementation of Bayesian regression models with alternative exposure metrics
- Model selection identifying best-fitting exposure metrics overall and by setting
- Analysis of household co-participation effects on transmission risk

Prerequisites and skills:

- Programming skills (R or Python required)
- Understanding of regression models

References

[1] Fairbanks EL, Bolton KJ, Jia R, Figueredo GP, Knight H, Vedhara K. (2023). *Influence of setting-dependent contacts and protective behaviours on asymptomatic SARS-CoV-2 infection amongst members of a UK university*. Epidemics. 43:100688. https://doi.org/10.1016/j.epidem.2023.100688.

[2] Bolton KJ, McCaw JM, Forbes K, Nathan P, Robins G, Pattison P, Nolan T, McVernon J. (2012). *Influence of contact definitions in assessment of the relative importance of social settings in disease transmission risk*. PloS one. 7(2):e30893. https://doi.org/10.1371/journal.pone.0030893

[3] Gelman A, Jakulin A, Pittau MG, Su YS. *A weakly informative default prior distribution for logistic and other regression models*. (2008). The annals of applied statistics. https://doi.org/10.1214/08-AOAS191

[4] Wagenmakers EJ, Gronau QF, Dablander F, Etz A. (2022) *The support interval*. Erkenntnis. 87(2):589-601. https://doi.org/10.1007/s10670-019-00209-z

M6: Sensitivity analysis to determine data requirements for estimating mosquito demographic parameters

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Background

Mosquito lifespan directly influences their ability to transmit diseases, as only mosquitoes that survive long enough to become infectious and subsequently feed on multiple hosts contribute to pathogen transmission. However, mosquito lifespan cannot be directly observed in field settings. Instead, it must be inferred from indirect measurements in trap catch data, for example parity rates (proportion of females that have previously laid eggs). This is likely to have measurement error and sampling bias. Furthermore, mosquito behaviour often changes with ageâĂŤfeeding frequency, host preference, and resting location may all vary.

Description This project will involve using ODEs to model adult mosquito populations incorporating age structure and predict the observed data in mosquito trap catches. Questions the student could answer include:

- What field data are most informative for estimating mosquito lifespan? Use sensitivity analysis methods to determine which observable quantities provide the best estimation of adult lifespan.
- How do age-dependent feeding patterns and targeted interventions affect population demographics? Explore how interventions that differentially impact mosquitoes of different ages influence overall population structure and lifespan estimates.

Methods may include; compartmental modelling using ordinary differential equations, global sensitivity analysis (e.g., PRCC, SobolâĂŹs method) and simulation studies.

Deliverables

- Exploratory simulations of age-structured mosquito population model
- Implementation and comparison of sensitivity analysis frameworks
- Simulation study demonstrating how intervention targeting affects demographic inference

Prerequisites and skills:

- Programming skills (R or Python required)
- Knowledge of differential equations and numerical methods

References

[1] Keeling, M. and Rohani, P. (2008). *Modeling Infectious Diseases in Humans and Animals*. Princeton University Press. https://doi.org/10.1515/9781400841035 [e-copy can be downloaded from the library]

[2] Derek Charlwood J, Nenhep S, Sovannaroth S, Morgan JC, Hemingway J, Chitnis N, BriÃńt OJ. (2016). *âĂŸNature or nurtureâĂŹ: survival rate, oviposition interval, and possible gonotrophic discordance among South East Asian anophelines*. . Malaria Journal. 15(1):356. https://doi.org/10.1186/s12936-016-1389-0

[3] Tenywa FC, Musa JJ, Musiba RM, Swai JK, Mpelepele AB, Okumu FO, Maia MF. (2024). *Sugar and blood: the nutritional priorities of the dengue vector, Aedes aegypti*. Parasites & Vectors. 17(1):26. https://doi.org/10.1186/s13071-023-06093-5.

M7: Data-driven strategy for word games

Proposer: Neil Morrison
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Background

Word games often involve large amounts of data and high operational complexity due to vast numbers of initial conditions and possible sequences of moves. Strategies for 'solving' word games must be able to quantify a wide range of game states in terms of an objective function to be extremised over all possible moves, or sequences of moves, in order to choose the best move to play next. Various factors influence the overall complexity of any particular strategy, e.g. the depth of search, the size of the dictionary, the number of letters, and certain details of the rules of the game. Algorithmic implementations of a strategy must consider prudently which data structures to maintain and which to generate on the fly. Aspects of dynamic programming and surrogate optimisation are conducive to forming an effective and versatile strategy over the course of a game.

Description

This project involves computer programming to investigate strategic aspects of a word game with a high operational complexity, for example Bananagrams, Word Soup, Scrabble, Wordle. The choice of game and the nature of the data to be gathered and analysed will be discussed with the supervisor, as will the objectives of a computer program to 'solve' that game. The student will develop and deploy various strategies and calibrate them based on data generated over many trial games. Code will be benchmarked and, where possible, comparisons against gameplay by top human players will be made.

Deliverables

- a computer program that can compete with top human players in a particular word game.
- a methodology by which to use gameplay data to refine an algorithmic strategy.

Prerequisites and skills:

Programming fluency is an essential pre-requisite for this project. Familiarize yourself with the games "Bananagrams" and "Word Soup." The latter can be installed on Android, Chromebook, Amazon Fire-stick, and on most computers via Android emulators such as BlueStacks & LeapDroid.

References

[1] D.P. Bertsekas, "Dynamic programming and optimal control," Belmont, 2005 (3rd edition).

[2] D. Bertsimas and A. Paskov, "An Exact Solution to Wordle," Operations Research, 2024.

[3] A.W. Appel and G.J. Jacobson, "The World's Fastest Scrabble Program," Communications of the ACM, Vol. 31 No. 5, 1988.

[4] P. Helling, A. Roy, and A. Olmsted, "Effects of Lexicon Size on Solving Bananagrams," 12th International Conference for Internet Technology and Secured Transactions, 2017.

M8: Analysis of the Urban Heat Island in Manchester

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Background

More than half of the world's human population lives in urban areas and it is expected that the global rate of urbanization will increase to around 70% by 2050[1]. Global warming and pollution are some of the negative impacts of urbanization and the cumulative effect of these is the Urban Heat Island (UHI). The UHI describes urban areas that are hotter than nearby rural areas. The US Environmental Protection Agency (EPA) reported that, although annual mean air temperature of a city with 1 million people or more can be just $1 - 3°\mathrm{C}$ warmer than its surroundings, on a clear, calm night, the difference can be as high as $12°\mathrm{C}$. UHI can affect communities by increasing summertime peak energy demand, air conditioning costs, air pollution and greenhouse gas emissions, heat-related illness and mortality, and water quality. It's no surprise that a great amount of work has been done in the past decades to investigate the effects of UHI in more depth.

Description

Data has been collected over several years to investigate the UHI effect in Greater Manchester, by setting up fixed point monitoring stations over the city. The data can be enhanced by considering other online data sources, such as the Met Office [2] and World Weather Online (WWO) [3]. The main aim of the project is to investigate various approaches to model the Manchester UHI data. The student is expected to review and apply to the data, different statistical approaches proposed in the literature. Possible choices for methods could be longitudinal, times series or spatio-temporal models. The application of the different procedures could be carried out in R or Python.

References

[1] United nations https://www.un.org/uk/desa/68-world-population-projected-live-urban-areas-2050-says-un

[2] Met Office https://www.metoffice.gov.uk/

[3] World Weather Online https://www.worldweatheronline.com/greater-manchester-weather/gb.aspx

M9: Efficient multiclass classification with feed-forward neural networks

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proposer: David Silvester
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Background

Shallow feed-forward neural networks have been developed that reliably classify images, such as those in the MNIST dataset. The computational efficiency of shallow feed-forward networks is, however, highly dependent on algorithmic details such as initialisation of weights, the choice of optimisation strategy and the vectorisation of key components.

Description

The purpose of this project is to explore the efficiency of classification algorithms that have been coded from scratch in a Python environment with linear algebra components taken from the NumPy package. The starting point will be to develop code to classify the digits from 1 to 99 into 4 distinct groups (the Fizz Buzz interview problem). Coding from scratch will facilitate a critical evaluation of the efficiency of the components of the algorithmic implementation. Timing comparisons with PyTorch classification results will be used to benchmark the efficiency of the developed code. The suggested starting point for code development is the book written by Joel Grus.

Deliverables

- A suite of hand-coded python routines.
- A detailed description of a selection of the computational experiments.

Prerequisites and skills:

This project requires a working knowledge of the mathematics underlying neural network components (activation functions, back propagation, gradient descent, Adam acceleration and logistic regression).

References

[1] MNIST database

[2] Fizz Buzz interview problem statement

[3] Joel Grus. *Data Science from Scratch.* O'Reilly Media, second edition, 2019. ISBN: 9781492041139.

M10: Explainable Clustering Methods

Proposer: Stefan Güttel
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Background

Clustering is an important unsupervised machine learning method to group nearby data points together while assigning distinct data points into separate groups (clusters). Among the many algorithms available, only a few methods are able to "explain" their outputs. The methods are usually referred to as "explainable" or "interpretable".

Description

The purpose of this project is to describe, analyse, and compare explainable/interpretable clustering algorithms. There are a number of reviews that can serve as a starting point [1, 2, 3], as well as concrete methods with some software implementations readily available [4, 5, 6].

Deliverables

- A concise and unified review of selected existing explainable clustering methods
- Mathematical analysis of some aspects of these methods, which could focus, e.g., on convergence properties or numerical stability issues
- Benchmarking of selected methods on some provided clustering datasets, performance comparison and interpretation

Prerequisites and skills: This project requires a good knowledge of mathematics, numerical computing, and some programming skills (Python or MATLAB).

References

[1] Hu, L., Jiang, M., Dong, J., Liu, X., & He, Z. (2024). *Interpretable clustering: A survey.* arXiv preprint arXiv:2409.00743.

[2] Dewoprabowo, R., Stefanus, L. Y., & Saptawijaya, A. (2025). *Explainable clustering: Methods, challenges, and future opportunities.* Journal of Intelligent Systems, 34(1), 20240477.

[3] Chen, J. (2018). *Interpretable clustering methods* (Doctoral dissertation, Northeastern University).

[4] Chen, X., and GÃijttel, S. (2024). *Fast and explainable clustering based on sorting.* Pattern Recognition, 150, 110298.

[5] Sabbatini, F., & Calegari, R. (2023). *Explainable clustering with CREAM.* In Proceedings of the International Conference on Principles of Knowlegde Representation and Reasoning (pp. 593–603).

[6] Moshkovitz, M., Dasgupta, S., Rashtchian, C., & Frost, N. (2020, November). *Explainable k-means and k-medians clustering.* In International Conference on Machine Learning (pp. 7055–7065).

M11: Fast Ridge Regression via Krylov Subspace Solvers

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Background

Ridge regression is widely used in machine learning, statistics, and data science for handling illconditioning and high-dimensional feature spaces. Classical implementations often rely on solving the associated normal equations using dense factorizations (e.g. Cholesky, QR, or SVD). While robust, these approaches can become prohibitively slow or memory-intensive for very large datasets or for matrices with millions of features.

Iterative Krylov subspace methods (such as Conjugate Gradient, LSQR, and LSMR) offer an appealing alternative in these settings. They avoid forming $X^T X$, rely only on matrix-vector products, and can exploit sparsity. Their performance, however, depends critically on properties of the data and the choice of preconditioner, making it non-trivial for practitioners to know when iterative methods outperform direct techniques.

Description

This project investigates efficient, matrix-free iterative solvers for the ridge regression system

$$\left(X^T X + \lambda I\right) \beta = X^T y,$$

and aims to develop practical guidance on when these solvers offer advantages over standard direct methods. The work includes:

- building functional wrappers for Krylov subspace solvers such as CG, LSQR, and LSMR using scipy.sparse.linalg,
- implementing matrix-free products with X and X^T;
- adding logging of residuals, iteration counts, and wall-clock times,
- evaluating lightweight preconditioners, such as diagonal/Jacobi preconditioning or small sketchbased approximations (optional, times-permiting);
- building a reproducible benchmarking framework for large-scale datasets;
- applying the methods to real datasets (e.g. text classification features, high-dimensional tabular data, recommender-system matrices) to quantify speed, accuracy, and memory performance;
- generating user-oriented recommendations on when iterative solvers outperform standard machinelearning libraries.

The project blends numerical linear algebra, scientific programming, and applied data analysis.

Deliverables

- A clear, well-documented implementation of CG, LSQR, and LSMR for ridge regression, including matrix-free operators and optional preconditioning.
- A benchmarking suite that measures accuracy, residual decay, runtime, and memory usage across synthetic and real datasets.
- A comparative study of direct vs. iterative solvers, with quantitative results and visualisations.
- A real-data analysis component using at least two substantial datasets, demonstrating practical performance differences.
- A written MSc thesis summarising methodology, experiments, and recommendations for practitioners.

Prerequisites and skills:

- Good programming skills in Python (NumPy/SciPy; familiarity with scikit-learn is helpful).
- Basic knowledge of linear algebra; familiarity with iterative methods is an advantage but not required.

References

[1] Benzi, Michele. *Preconditioning Techniques for Large Linear Systems: A Survey*, Journal of Computational Physics (2002) Vol. 182, No. 2, pp. 418-477, doi:10.1006/jcph.2002.7061.

[2] Golub, Gene H., and Van Loan, Charles F. (2013) *Matrix Computations*, Society for Industrial and Applied Mathematics.

[3] Hastie, Trevor. *Ridge Regularization: An Essential Concept in Data Science*, Technometrics (2020) Vol. 62, No. 4, pp. 426-433, doi:10.1080/00401706.2020.1791959.

[4] Hastie, Trevor and Tibshirani, Robert and Friedman, Jerome. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, second edition (2009), Springer.

[5] Hoerl, Arthur E. and Kennard, Robert W. *Ridge Regression: Biased Estimation for Nonorthogonal Problems*, Technometrics (1970) Vol. 12, No. 1, pp. 55-67, doi:10.1080/00401706.1970.10488634.

M12: Improving Tests for Circadian Disruption

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Background

The circadian clock is an evolutionarily highly conserved endogenous timing program regulating physiology in response to time of day. Disruption of this mechanism is associated with poor patient outcomes, including increased patient mortality. We know how to treat this condition, however the condition is difficult to detect. This project will focus on developing methods to detect the condition, thereby potentially improving patient outcomes.

Description

We have already developed a method that can identify circadian disruption from 6 samples however this is difficult to use due to the number of samples required. The intention of this project is to develop a method which uses fewer samples. This will involve use of machine learning to map the multidimensional state of multiple oscillatory processes. Initial work has been encouraging, but inconclusive. The intention of this project is to examine whether the inclusion of additional biological inforamation about the core clock oscillator will lead to a viable method.

Deliverables

- Develop novel methods that can identify circadian disruption in patients.
- Provide implementation of the methods in open-source software.
- Analyse data to assess the clinical effects of circadian disruption.

This project has the potential to lead to scientific publications.

Prerequisites and skills:

Good understanding of mathematics especially machine learning; Ability to use SQL and R; Good data management practices.

References

[1] P.S. Cunningham (2023) *et al ClinCirc identifies alterations of the circadian peripheral oscillator in critical care patients*. In *J. Clin. Invest.* 133(4) e162775 DOI: 10.1172/JCI162775.

M13: The Probability of the Unseen: The Good-Turing Estimator

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Background

The Good-Turing estimator is a statistical technique developed by A. Turing and I.J. Good in the 1940s to estimate the probability of encountering unseen species or events based on past observations. This methods helps improve the accuracy of probability estimates, especially for rare occurrences, by adjusting the observed frequencies of events.

The Good-Turing method has applications in many diverse fields such as ecology, linguistics, genetics and cryptography, and is closely linked to empirical and nonparametric Bayes.

Description

In this project you will:

- study the mathematical and statistical underpinnings of the Good-Turing method,
- explore todayâĂŹs relevance of the Good-Turing estimator in statistics and machine learning, and
- apply it to relevant modern data (e.g. in ecology or genetics).

Deliverables

- A thorough overview of the theory underlying Good-Turing estimator (including a derivation from first principles if the focus of the thesis is on theory).
- Numerical experiments studying the properties of the Good-Turing estimator (including a comparison with competing methods if the focus of the thesis is methodological).
- Application of the estimator to real data, e.g. identification of rare cancer variants (multiple data sets will be explored and interpreted if the focus of the thesis is applied).

Prerequisites and skills:

This project is suitable for multiple students and can be adapted to either focus more on theory, on methods or on the practical applications of the Good-Turing estimator, depending on the interest of the student.

Good understanding of statistical estimation and inference methods is required, familiarity with Bayesian statistics is a plus.

References

[1] Chakraborty, S., et al. 2019. *Using somatic variant richness to mine signals from rare variants in the cancer genome.* Nat. Comm. **10**:5506. https://doi.org/10.1038/s41467-019-13402-z

[2] Chao, A., et al. 2017. *Seen once or more than once: applying GoodâĂŞTuring theory to estimate species richness using only unique observations and a species list.* Meth. Ecol. Evol. **8**:1221–1232. https://doi.org/10.1111/2041-210X.12768

[3] Favaro, S., et al. 2015. *Rediscovery of GoodâĂŞTuring estimators via Bayesian nonparametrics.* Biometrics **72**:136–145. https://doi.org/10.1111/biom.12366

[4] Orlitsky, A., and Suresh, A.T. 2015. *Competitive distribution estimation: why is Good-Turing so good?* Proc. 29th Intl. Conf.Neural Inf. Proc. Syst. 2143–2151 https://dl.acm.org/doi/10.5555/2969442.2969479

[5] Robbins, H. E. 1968. *Estimating the total probability of the unobserved outcomes of an experiment.* Ann. Math. Statist. **39**:256–257 https://doi.org/10.1214/aoms/1177698526

[6] Good, I. J. 1953. *The population frequencies of species and the estimation of population parameters.* Biometrika **40**:237–264 https://doi.org/10.1093/biomet/40.3-4.237

M14: Bayesian reconstruction of binary images using the Ising model

Proposer: Jonas Latz
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Background

A binary image is given by a matrix $I \in \{0, 1\}^{n \times m}$. Binary images have a multitude of applications, they are frequently used to represent texts or after a grayscale or colour image has been segmented. It is not uncommon that such images contain noise, which, in this case, is a random flip of some of the pixels.

Description

The purpose of this project is to develop a Bayesian methodology to reduce noise in binary images using a Bayesian methodology based on an Ising prior. The Ising prior is a probability distribution on a lattice that is popularly used in statistical mechanics to represent ferromagnetism. It will be able to represent local features in the binary image while smoothing the noisy pixels.

Deliverables

- A derivation of the posterior distribution of the image based on Bernoulli noise and Ising prior using BayesâĂŹ formula.
- Implementation of a Gibbs sampler in 2D that can sample from the posterior derived previously.
- Testing of different methodologies to make the Gibbs sampler more efficient: especially tempering; possibly data augmentation.
- Finding ways to scale the methodology to large images.

Prerequisites and skills:

The project requires an excellent understanding of theory and application of Markov chain Monte Carlo methods, for instance, based on the course 'Introduction to Uncertainty Quantification'. Students without an excellent background in Markov chain Monte Carlo will not be able to take this project. Programming one of {Matlab, Python, R, C, C++, Julia} will be necessary.

References

[1] Robert, Christian P., and George Casella (2004) *Monte Carlo Statistical Methods*, Springer.

M15: Fake news detection

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Background

With the rapid development and popularization of Internet technology, the propagation and diffusion of information become much easier and faster. While making life more convenient, the Internet also promotes the wide spread of fake news. Fake news is not a recent concept, but it is a commonly occurring phenomenon in current times. The consequence of fake news can range from being merely annoying to influencing and misleading societies or even nations. Thus it is of utmost importance for individuals and societies to be able to judge the credibility of it.

Description

A lot of research efforts have been made to combat fake news and a variety of approaches exist to identify fake news. By conducting a systematic literature review, we will identify the main approaches currently available to identify fake news and how these approaches can be applied in different situations. Some approaches are illustrated with a relevant example as well as the challenges and the appropriate context in which the specific approach can be used.

Deliverables

- Review of the relevant literature on methods for fake news detection, including statistical, machine learning, and deep learning models.
- Choose publicly available data sets.
- Implement and compare models such as logistic regression, random forests, support vector machines, and deep learning.
- Assess performance using accuracy, precision, recall, F1âĂŚscore, and ROCâĂŚAUC.
- Explore explainability to understand model decisions.

Prerequisites and skills:

- Good command of programming (preferably in R).
- Ability to understand classification algorithms.
- Inclination to learn advanced algorithms.

References

[1] Bo Hu, Zhendong Mao, Yongdong Zhang. *An overview of fake news detection: From a new perspective,* . Fundamental Research. 5(1), 332-346, 2025.

[2] de Beer, D., Matthee, M. *Approaches to Identify Fake News: A Systematic Literature Review*. In: Antipova, T. (eds) Integrated Science in Digital Age 2020. ICIS 2020. Lecture Notes in Networks and Systems, 136. Springer, Cham.

[3] Xinyi Zhou, Reza Zafarani *A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities*. ACM Computing Surveys (CSUR). 53(5) 1-40, 2020.

M16: Bootstrapping with large data

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Background

The bootstrap provides a simple and powerful means of assessing the quality of estimators. However, in settings involving large data sets, which are increasingly prevalent, the calculation of bootstrap-based quantities can be prohibitively demanding computationally.

Description

However, the original version of bootstrap method may not be computationally feasible in the context of large datasets, as repeated resampling from the entire data may not even be feasible given the computational facilities available. Thus improvements to the bootstrap method have been made in recent years, wherein they assess the quality of estimators by subsampling the full dataset before resampling the subsamples. A few such alternate methods that have been proposed in literature, are "m out of n bootstrap", "bag of little bootstraps", "optimal subsampling" to mention a few.

The performance of these recent subsampling methods is naturally influenced by tuning parameters such as the size of subsamples, the number of subsamples, and the number of resamples per subsample.

In addition the performance may also be context dependent, that is the complexity of the estimation procedure itself may have a bearing on the outcome of these procedure.

Thus it is worth investigating which of these methods are, firstly, scalable in reality when data is really large. Also may be useful to understand how they cope with complexity of the problem.

Deliverables

- Review of the relevant literature on large data bootstrap.
- Choose multiple algorithms.
- Simulations run across multiple (real and/or artificial) datasets,
- Assess their performances over growing size.
- For simulated data one can also assess effect of complexity of estimation problem and it's effect on "performance".

Prerequisites and skills:

- Programming skill in R software is highly desired, although python also could be used.
- Nevertheless to assess performances of existing statistical algorithms R would be the preferred platform.
- Requires to be familiar with simulation.
- Some degree of exposure in Statistical computing regarding applications of bootstrap methods would help.

References

[1] Bickel, P. J., Gotze, F. and van Zwet,W.. *Resampling fewer than n observations: gains, losses, and remedies for losses*. Statist. Sin., 7, 1 – 31, 1997.

[2] Kleiner A, Talwalkar A, Sarkar P and Jordan MI. *A scalable bootstrap for massive data*. J. R. Statist. Soc. B, 76, Part 4, pp. 795 – 816, 2014.

[3] Ma Y, Leng C and Wang H. *Optimal Subsampling Bootstrap for Massive Data*. Journal of Business & Economic Statistics, DOI:10.1080/07350015.2023.2166514, 2023.

M17: Dictionary Learning with applications

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Background

Dictionary Learning (DL) is a feature learning method that derives a finite collection of dictionary elements (atoms) from a given dataset. These atoms are small characteristic features representing recurring patterns within the data. A dictionary therefore is a compact representation of complex or large-scale datasets.

Description

While deep learning is most commonly used as a supervised learning scheme for classification tasks, DL is a mostly unsupervised learning approach that aims to find a set of features permitting sparse representation of the data at hand. Such a sparse representation (or sparse code) allows reconstructing, with some degree of error ϵ, any complex signal with a linear combination of few features. Such a sparse representation of complex signals holds significant potential in signal decomposition where one wants to find a small set of signal components that describe underlying patterns and therefore permit separate analysis and processing of different types of patterns. See Dumitrescu(2018) for various DL methods their applications.

A particular application area that is less investigated however could be of high utility is DL of temporal data. In this project we will review literature available on this topic and explore implementation techniques.

Deliverables

- A review of the relevant literature.
- Review of implementations of DL available in various packages.
- Implementation of DL (preferably on temporal/time-series) data.
- Time permitting work on real data set suited to type of learning.

Prerequisites and skills:

- Good command of programming (preferably in R).
- Ability to understand supervised/unsupervised algorithms.
- Inclination to learn advanced algorithms.

References

[1] Lewicki MA, and Sejnowski TJ. *Learning Overcomplete Representations*. Neural Computation, vol. 12, no. 2, pp. 337 – 365, 2000

[2] Dumitrescu B and Irofti P. *Dictionary Learning Algorithms and Applications*. Springer. https://doi.org/10.1007/978-3-319-78674-2, 2018.

[3] Burger J and Calvimontes J . *Temporal Dictionary Learning for time-series decomposition*. Investigacion & Desarrollo 19(1):105-112. DOI: 10.23881/idupbo.019.1-7i, 2019.

M18: Algorithm Bias and Data Science Ethics

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Background

Data-driven machine learning systems learn from hidden patterns in existing data and risk reproducing and amplifying social patterns of bias reflected in this data, such as racial or gender bias. Algorithmic fairness is an emerging research area for removing these wrongful societal biases in the hope of making data-driven AI systems ethical.

Description

In recent years a significant amount of research has been done attempting to define fairness quantitatively. In particular, defining fairness in the context of decisions based on the predictions of statistical and machine learning models.

Though the algorithmic fairness conversation is somewhat new, it resembles older work. For example, since the 1960s, psychometricians have studied the fairness of educational tests based on their ability to predict performance.

This field is new and naturally suffers from drawbacks like inconsistent motivations, terminologies, and notations. Thus it is not any easy task to understand the literature in a comparative sense.

Thus in this project we would simplify the matters and look at the problem as a potentially sampling bias issue. We will consider some well-established predictive algorithms. Assume and or define "fairness" in prediction. By varying sampling schema we would like to assess the susceptibility of the chosen algorithms in predicting (un-)fairly.

Deliverables

This project has number of possibilities as outcomes. The method outlined above of choosing a set of algorithms and assessing them via simulation could involve the followings

- Review of the relevant literature.
- Simulations run across multiple (real and/or artificial) datasets.
- Use predetermined (single or multiple) definition(s) of "fairness".
- Assess "fairness" performance of the algorithms for various data sets (and also possibly under various definitions of fairness).

Prerequisites and skills:

- Programming skill in R software is highly desired, although python also could be used. Nevertheless to assess performances of existing statistical algorithms R would be preferred platform.
- Requires to be familiar with simulation.
- Some degree of exposure in predictive algorithm would help.

References

[1] Luciano Floridi. *The Ethics of Artificial Intelligence: Principles, Challenges, and Opportunities*. OUP, 2023.

[2] Shira Mitchell, Eric Potash, Solon Barocas, Alexander D'Amour, Kristian Lum. *Algorithmic Fairness: Choices, Assumptions, and Definitions*. Annual Review of Statistics and Its Application 8:1, 141-163, 2021.

[3] Jeff Larson, Surya Mattu, Lauren Kirchner and Julia Angwin. *Machine Bias*. Pro Publica, 2016.

M19: Measuring Performance of a code in R: Large data context

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Background

R programming language is one of the most excellent facilities available to today's data scientist and is known to possess Turing completeness. It is quite versatile in its ability to cater to various data domains like API's, dashboards, apps and much more. But as with some other such languages, it takes practice to be able to write a "Production Level Code" in R.

Description

In this project we will focus on only one such aspect that is desired from any successful production level code, that is its performance, with regard to speed, memory, resource requirement. Such an assessment becomes critical especially in the context of large data. Firstly we will have to understand how to benchmark the performance of an R code. This also could be assessed from various perspectives, for example in I/O part of the code, or algorithm part(s) of the code, etc. This will necessitate learning about resources like "microbench", "tidyverse" etc that are available within R that help us assess and improve our codes.

Deliverables

This project has number of possibilities as outcomes. A few simple algorithms on large data could be attempted. Else complex algorithms on medium sized data could also be explored.

- Review of the relevant literature is essential to avail latest programming facilities available in R.
- Many medium sized data sets are already available within R, including benchmark data, which could be used. It is also possible to simulate medium (and complex) or large data sets.
- Literature review to define and quantify "performance".
- Implementation of the above in R environment and discussion of findings.

Prerequisites and skills:

- Programming skill in R software is essential.
- Familiarity with standard (and competing) algorithms available in R for solving standard problems (e.g. clustering, classification, regression, etc) will be advantageous.
- It is encouraged to have own data and algorithm of interest.

References

[1] *Turing Completeness*. “https://en.wikipedia.org/wiki/Turing_completeness”, (Last accessed 24-Nov-2023).

[2] Colin Gillespie, Robin Lovelace. *Efficient R Programming*. O’Reilly Media, Inc. ISBN: 9781491950784, 2016.

[3] Aloysius Lim, William Tjhi. *R High Performance Programming*. Packt Publishing. ISBN 9781783989270, 2015.

M20: Data efficiency of algorithms: Survey and comparison

Proposer: Madhu Bhattacharjee
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Background

The key advancements that have enhanced the growth of AI are 1) sufficiently powerful processors, 2) parallelizing the deep learning algorithms and importantly 3) availability of big data upon which we implement the algorithms. It has also been recognised that for many methods performance suffers greatly on small data.

Description

We aimed to study the predictive performance of different modelling techniques in relation to the effective sample size ("data hungriness"). In statistics, the topic was known under the name of (asymptotic) relative efficiency and was of great interest. There's extensive literature on efficiency of many parametric model-based estimators and testing procedures.

With availability of massive data, researchers are inclined to draw insights where one is not even sure how to formulate the hypothesis ahead of time. Along with that there is general trend not to dwell on efficiency of the methods/algorithms with regard to data (i.e. sample size).

Thus we would develop a systematic scheme to define learning curves that reflect the performance of an algorithm (or model) in terms of predictive ability. It is common to expect that such an ability of a would be a monotonically increasing function of the sample size, converging to a maximum at the infinite sample size. But recent exercises have shown that sample size alone may not be the dictating factor and increase in (dependence or) complexity may also contribute to (lack-of) efficiency.

There are multiple alternate hypotheses that have been explored in literature, for example modern (and more) flexible techniques are more "data hungry" than more conventional modelling techniques, such as regression analysis, supervised techniques are trained on labelled data thus they are more data hungry than unsupervised methods, etc.

Deliverables

- Review of the relevant literature on data hungriness.
- Choice of algorithms with justifications.
- Simulations run across multiple (real and/or artificial) datasets,
- Assess their performances over growing sample size.
- For simulated data one can also assess effect of complexity of background data and it's effect on "performance".

Prerequisites and skills:

- Programming skill in R software is highly desired, although python also could be used.

- Requires to be familiar with simulation and data analysis.
- Some degree of exposure in predictive algorithm would help.

References

[1] Hallin M. *Asymptotic Relative Efficiency: Theory*. Encyclopedia of Environmetrics, John Wiley & Sons, 2013.

[2] Tzoulaki I, Liberopoulos G, Ioannidis JPA. *Assessment of claims of improved prediction beyond the Framingham risk score*. JAMA, 302:2345 – 2352, 2009.

[3] Lee DB. *Requiem for large-scale models*. ACM SIGSIM Simul Dig, 16 – 29, 1975.

[4] van der Ploeg T, Austin PC, and Steyerberg EW. *Modern modelling techniques are data hungry: a simulation study for predicting dichotomous endpoints*. BMC Medical Research Methodology, 14:137, 2014.

M21: On the Chemical Basis of Morphogenesis

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Background

It is suggested that a system of chemical substances, called morphogens, reacting together and diffusing through a tissue, is adequate to account for the main phenomena of morphogenesis.

Description

The purpose of this project is to explore a possible mechanism by which the genes of a zygote may determine the anatomical structure of the resulting organism. The theory does not make any new hypotheses; it merely suggests that certain well-known physical laws are sufficient to account for many of the facts.

Deliverables

- A suite of illustrative examples
- Numerical experiments about the patterning of animal coats.

Prerequisites and skills:

The full understanding of the project requires a good knowledge of mathematics, some biology, and some elementary chemistry. Since students cannot be expected to be experts in all of these subjects, a number of introductory books and articles are suggested below.

References

[1] Fraser, Andrew M (2008) *Hidden Markov models and dynamical systems*, Society for Industrial and Applied Mathematics.

[2] Bhar, Ramaprasad, and Shigeyuki Hamori (2004) *Hidden Markov models: applications to financial economics*, Springer Science and Business Media.

[3] Y Zhao and H Zhang. *Investigating the inter-country variations in game interruptions across the Big-5 European football leagues.* International Journal of Performance Analysis in Sport 21.1 (2021), pp. 180âĂŞ196. DOI: 10.1080/24748668.2020.1865688.

M22: Physics Informed Neural Networks for the Landau-de Gennes theory for Nematic Liquid Crystals

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Proposer: Professor Apala Majumdar (up to 3)
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Background

Liquid crystals (LCs) are partially ordered materials that combine fluidity with the ordering characteristics of solids. The simplest LC phase is the nematic liquid crystal (NLC) phase for which the constituent rod-like molecules move freely but tend to align along certain locally preferred directions referred to as "directors" in the literature, and these directors define the orientational/directional ordering of NLC phase. Consequently, NLCs have anisotropic or direction-dependent physical, optical, and mechanical properties , making them the working material of choice for electro-optic devices, photonics, functional materials and some healthcare technologies .

Mathematical modelling can play a crucial role in predicting and controlling the performance of NLC-based devices. We focus on the Landau-de Gennes theory for NLCs wherein the NLC phase is modelled by a macroscopic order parameter: the $\mathbf{Q}$-tensor order parameter, which is a symmetric, traceless 3×3 matrix that can be related to experimental observables. The Landau-de Gennes free energy is a nonlinear and non-convex functional of the $\mathbf{Q}$-tensor and its spatial derivatives, subject to the imposed boundary conditions. The Landau-de Gennes energy minimizers model the physically observable configurations and are hence, of practical interest. The non-energy minimizing critical points of the Landau-de Gennes energy control the dynamics of a NLC system. Collectively, the energy minimizing and non-energy minimizing critical points of the LdG energy dictate the equilibrium and non-equilibrium properties of confined NLC systems, which are of technological significance. This necessitates the development of robust computational frameworks for the analysis of the Landau-de Gennes energy, as will be explored in this project.

Description

The main objective of this project is to implement physics-informed neural networks (PINN) that can accurately and efficiently compute the critical points of the Landau-de Gennes free energy for benchmark problems, e.g., on two-dimensional polygonal domains and three-dimensional cuboids, subject to experimentally relevant boundary conditions. The critical points of the Landau-de Gennes energy are classical solutions of the Euler-Lagrange equations which are a system of five, nonlinear and coupled partial differential equations (PDEs). We will employ PINN to solve the system of Euler-Lagrange equations associated with the Landau-de Gennes energy. PINN offer the possibility of automated algorithms for solving large and complex systems of PDEs in a domain-free and mesh-free manner, making them highly flexible and suitable for high-dimensional problems and complex geometries. We will compare PINN outputs with the results from classical numerical solvers, such as the finite difference method, accompanied by error estimates for PINN-based methods that can be used for refining and adapting PINN for complex liquid crystal problems.

Deliverables

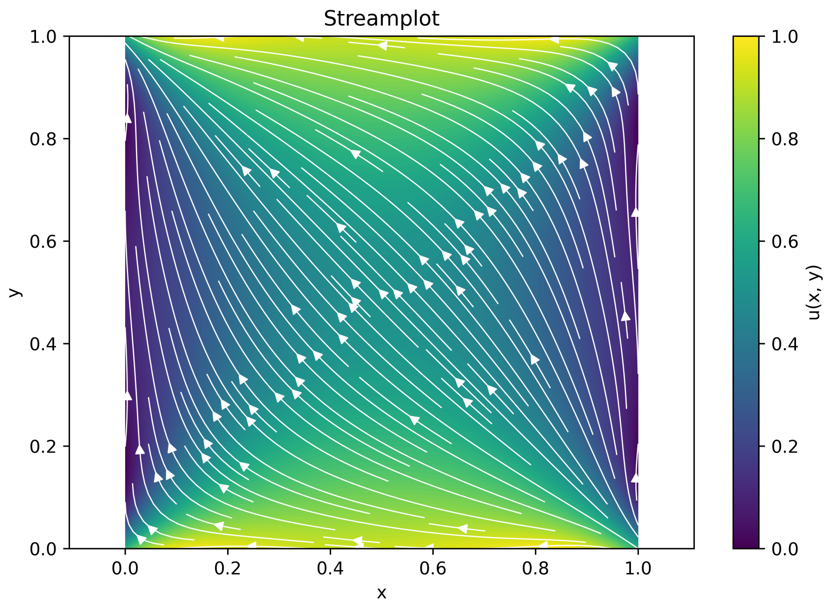

Figure 2: The PINN *diagonal* solution for NLCs on a square domain, with tangent boundary conditions. The trained model achieves a mean squared error of $\mathrm{MSE} = 2.94 \times 10^{-3}$, indicating high accuracy.

- Numerical code for PINN trained by data from classical numerical solvers.
- Numerical experiments on the application of PINN to NLCs in square domains, hexagons and cuboids as experimentally relevant gemetries.

Prerequisites and skills: This project requires a good knowledge of mechanics, partial differential equations and numerical methods for differential equations. Some references are given below.

References

[1] P. G. De Gennes and J. Prost. The physics of liquid crystals. Number 83. Oxford University Press, 1993.

[2] I. W. Stewart. The static and dynamic continuum theory of liquid crystals: a mathematical introduction. CRC Press, 2019.

[3] J. P. F. Lagerwall and G. Scalia. A new era for liquid crystal research: Applications of liquid crystals in soft matter nano-, bio- and microtechnology. Current Applied Physics, 12(6):1387âĂŞ1412, 2012.

[4] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational physics, 378:686âĂŞ707, 2019.

M23: Dynamic alternative routing of calls in loss networks

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Background

We consider an algorithm for routing calls in a communication network, first introduced and popularised by Frank Kelly and co-authors in the 1990s.

The network has N nodes, and is fully connected, meaning there is a link between each pair of nodes. Each link is assumed to have a capacity C, where C is a positive integer. In fact, we consider a sequence of these networks, one for each value of N and we assume the network size N is large.

Calls arrive into the network at rate $\lambda\binom{N}{2}$, where $\lambda > 0$ is a constant, and each call chooses a pair of endpoints uniformly at random. Each newly arriving call is routed directly between the two nodes if possible, that is if there are fewer than C calls occupying the direct link at the time of arrival. The call will then take up one unit of capacity on the link.

If this is not possible (that is the direct link is full), then an alternative route via one intermediate node are inspected (that is, a two-link path between the endpoints of the calls), and the call is routed along that route, if possible, that is if there is spare capacity on both links on the path. If the two-link path inspected is not available, then the call is lost.

Calls have independent exponentially distributed durations with mean 1 ; upon departure, the call vacates one unit of capacity on each link of the path it has been using.

When the size N of the network is large, the behaviour of this model can be well approximated by the solution to a system of differential equations.

The model exhibits a variety of different behaviours in the long-term. For certain parameter values, the equilibrium distribution of the numbers of links with different loads will concentrate tightly around the unique fixed point of the limiting system of differential equations. For other values of the parameters, the limiting differential equations have multiple fixed points, and the process exhibits metastability in the long-term.

Other versions of this model have also been studied, for instance where each link has some capacity reserved for direct calls and the rest of its capacity reserved for alternatively-routed calls; and also versions where alternative routes consist of pairs of links chosen uniformly at random, without respecting network topology.

Description

The purpose of this project is to explore some of the variants of the Kelly alternative routing model computationally. The focus will be on: simulating the underlying continuous-time Markov chain; finding numerical solutions to the limiting system of differential equations; computationally verifying the validity of the approximation of the Markov chain by the differential equation when the network size N is large; long-term behaviour of the differential equations and the Markov chain.

Deliverables

- Numerical study of the Markov process and its limiting differential equation for a variant of the Kelly dynamic alternative routing algorithm.

Prerequisites and skills:

- Strong programming skills, including in simulating stochastic processes and numerical solutions of differential equations;
- Basic knowledge of Markov chains.

References

[1] G. Brightwell and M. Luczak (2013). A fixed-point approximation for a routing model in equilibrium. Unpublishd, arXiv:1306.5002.

[2] J.P. Crametz and P.J. Hunt (1991). A limit result respecting graph structure for a fully connected loss network with alternative routing. Ann. Appl. Probab. 1436-444.

[3] R.J. Gibbens, P.J. Hunt, and F.P. Kelly (1990). Bistability in communication networks. In Disorder in Physical Systems (G.R. Grimmett and D.J.A. Welsh, eds.) 113-128, OUP.

[4] C. Graham and S. MÃl'lÃl'ard (1993). Propagation of chaos for a fully connected loss network with alternate routing. Stochastic Process. Appl. 44 159-180.

[5] B. Hajek (1987). Average case analysis of greedy algorithms for Kelly's triangle problem and the indepen dent set problem. 26th IEEE Conference on Decision and Control 26 1455-1460.

[6] F.P. Kelly (1991). Loss networks. Ann. Appl. Probab. 1 319-378.

[7] F.P. Kelly and E. Yudovina (2014). Stochastic networks. CUP.

[8] M.J. Luczak (2012) A quantitative differential equation approximation for a routing model. Unpublished, arXiv:1212.3221.

[9] M.J. Luczak and C. McDiarmid (2015). Balanced routing of random calls. Annals of Applied Probability25 1279-1324.

[10] M.J. Luczak, C. McDiarmid and E. Upfal (2003). On-line routing of random calls in networks. Probab. Theor. Relat. Fields 125457-482.

[11] M.J. Luczak and E. Upfal (1999). Reducing network congestion and blocking probability through balanced allocations. Proc. 40th IEEE Symp. Found. Comp. Sci. 587-595.

[12] D. Martirosyan and P. Robert (2020). The equilibrium states of large networks of Erlang queues. Advances in Applied Probability 52 617-654.

[13] J.R. Norris (1998). Markov chains. CUP.

[14] S. Olesker-Taylor (2022). Metastability in loss networks with dynamic alternative routing. Annals of Applied Probability 32 1362-1399.

M24: Long-Term Forecasting of Precipitation Extremes Using Gaussian Processes and Machine Learning Models

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Background

Extreme precipitation events, such as intense rainfall and prolonged wet periods, pose significant risks to society, including flooding, infrastructure damage, and agricultural disruption. With the increasing variability associated with climate change, understanding and forecasting such extremes has become a pressing scientific and engineering challenge. The Copernicus Climate Data Store (CDS) provides high-resolution, globally consistent reanalysis datasets, including ERA5 and ERA5-Land, which contain detailed information on precipitation and related atmospheric variables. These datasets make it possible to build predictive models for both short-term and long-term rainfall behaviour, including rare but high-impact extremes. Statistical techniques such as Gaussian Processes (GPs) provide interpretable, probabilistic forecasts with mathematically principled uncertainty estimates. In contrast, modern deep learning architectures, such as LSTMs, Temporal Convolutional Networks (TCNs), and Transformer-based sequence models, offer powerful tools for learning complex non-linear temporal dependencies. Comparing these two modelling paradigms, particularly in the context of precipitation extremes, is an important open question in climate data science.

Description

The purpose of this project is to investigate and compare two different approaches to forecasting precipitation extremes using CDS datasets: Gaussian Process regression models and state-of-the-art machine learning methods for time series forecasting. The project will involve extracting suitable precipitation data from the Copernicus Climate Data Store, constructing temporal predictors, and implementing a suite of forecasting models. Gaussian Processes with a range of kernel structures (e.g. MatÃĺrn, Rational Quadratic, Spectral Mixture) will be evaluated alongside deep learning models such as LSTMs, TCNs, and optionally long-sequence Transformer models. A central part of the project will be the exploration of uncertainty quantification techniques, including GP posterior variances, Monte Carlo dropout, deep ensembles, and conformal prediction. The student will compare short-term (1–7 days ahead) and long-term (1–3 months ahead) forecasting performance, with a particular emphasis on predicting extreme precipitation events. Performance will be assessed using standard regression metrics, as well as metrics designed for extremes such as quantile loss, Brier scores, and coverage probabilities. The final part of the project will involve synthesising results to determine the strengths, limitations, and practical applicability of statistical vs. machine learning approaches in climate forecasting.

Deliverables

- A cleaned and reproducible dataset derived from ERA5 / ERA5-Land precipitation data.
- Implemented Gaussian Process models with appropriate kernel choices and sparse approximations.

- Implemented machine learning forecasting models (e.g. LSTM, TCN, Transformer).
- A comprehensive comparison of forecasting accuracy for short- and long-term horizons.
- An evaluation of uncertainty quantification methods for both statistical and ML models.
- A well-documented codebase and a written dissertation summarising methodology, results, and recommendations.

Prerequisites and skills:

A good working knowledge of Python and R is required, including familiarity with data manipulation. Some prior exposure to machine learning or statistical modelling will be helpful, but the project is designed so that missing background can be acquired during the work. Students with stronger mathematical backgrounds will be able to explore Gaussian Processes and uncertainty quantification methods in greater theoretical depth, while students oriented toward applied data science may focus on model implementation and evaluation. Interest in climate data analysis or environmental applications is beneficial but not required.

References

[1] Rasmussen, C. E., and Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

[2] Hersbach, H. et al. (2020). *The ERA5 Global Reanalysis*. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049.

[3] Salinas, D., Flunkert, V., Gasthaus, J., and Januschowski, T. (2020). *DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks*. International Journal of Forecasting.

M25: Data-Driven Multistate Survival Modeling for Tracking Clinical Disease Progression

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Background

Multistate survival models provide a powerful framework for analyzing complex, time-dependent processes in which individuals transition between multiple states over time. Unlike traditional survival analysis, which focuses on a single endpoint, multistate models capture sequential transitions and enable dynamic prediction [1] of future outcomes based on updated patient information. [2, 3] Dynamic prediction continuously updates the probability of future events as a patient progresses through different states. Its main purposes are to improve individualized risk stratification, support clinical decision-making, enhance prognostic accuracy, and identify key factors influencing transitions at different stages. [4, 5] For example, in post-transplant patients, the risk of severe complications or death may change depending on whether the patient is currently recovering, experiencing mild complications, or has relapsed. This project focuses on post-transplant patients using the European Society for Blood and Marrow Transplantation (EBMT) dataset, which contains longitudinal time-to-event information for each patient. We consider four clinically relevant states: (1) recovery without complications, (2) mild complications or manageable events, (3) severe complications or relapse, and (4) death. Multistate modeling allows the investigation of how covariates-such as age, donor-recipient match, prophylactic regimen, and transplant year-influence both the probability and timing of transitions. [6, 7] By leveraging EBMT's detailed longitudinal data, this project aims to develop and evaluate multistate survival models that track patient trajectories, enable dynamic risk prediction, and support data-driven clinical decision-making.

Description

This project aims to develop and implement a data-driven multistate survival framework to model and predict patient transitions across disease states. For this aim we propse a baseline transition-specific Cox models (refrence). Also, we propose dynamic prediction which is estimate patient-specific probabilities of being in each state at future time points. [5] This subject has potential for extending for more complex model in the future. [3]

Deliverables

- Numerical experiments analyzing transition hazards, state probabilities, and covariate effects.
- Dynamic prediction outputs showing patient-specific probabilities of future states over time.
- High-quality visualizations including transition diagrams, AalenâĂŞJohansen curves, and hazard plots.
- Reproducible R scripts and workflows for multistate modeling and dynamic prediction.

Prerequisites and skills:

Regression analysis, survival analysis, Programming.
Optional: Bayesian modeling for advanced extensions.

References

[1] Hashemi R, Baghfalaki T, Philipps V, Jacqmin-Gadda H. Dynamic prediction of an event using multiple longitudinal markers: a model averaging approach. Statistics in Medicine. 2025;44(1314):e70122.

[2] van den Hout A. Multi-State Survival Models for Interval-Censored Data. Chapman and Hall/CRC; 2017.

[3] Urbut SM, Yeung MW, Khurshid S, Cho SMJ, Schuermans A, German J, et al. MSGene: a multistate model using genetic risk and the electronic health record applied to lifetime risk of coronary artery disease. Nature Communications. 2024;15(1):4884.

[4] van Houwelingen HC, Putter H. Dynamic prediction in clinical survival analysis. Boca Raton: Chapman & Hall/CRC; 2012.

[5] van Houwelingen HC, Putter H. Dynamic predicting by landmarking as an alternative for multi-state modeling: an application to acute lymphoid leukemia data. Lifetime Data Analysis. 2008;14(4):44763.

[6] Fiocco M, Putter H, van Houwelingen HC. Reduced-rank proportional hazards regression and simulation-based prediction for multi-state models. Statistics in Medicine. 2008;27(25):4340-58.

[7] de Wreede LC, Fiocco M, Putter H. mstate: An R package for the analysis of competing risks and multi-state models. Journal of Statistical Software. 2011;38(7):1-30.

M26: Weighted Generalized Estimating Equations for Correlated Data with Missingness

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Background

Longitudinal and clustered data arise when multiple measurements are collected on the same subjects or units, leading to correlated observations. Standard regression methods assuming independence are inappropriate, motivating the use of Generalized Estimating Equations (GEE) [1] for robust populationaverage inference. Weighted Generalized Estimating Equations (WGEE) [2, 3, 4] extend classical GEE methods to handle missing data, dropout, and informative cluster sizes in such correlated datasets. Standard GEE may produce biased estimates if data are missing at random or if cluster size is associated with the outcome. WGEE incorporates observation-level or cluster-level weights, including inverse-probability or stabilized weights, to achieve consistent and efficient estimation. Recent developments include penalized WGEE, which allows variable selection and estimation in highdimensional datasets while maintaining desirable asymptotic properties, as illustrated by Ma et al. [5] GEE and WGEE is implemented in R via packages such as geepack making it practical for complex correlated datasets with missingness. Also, PGEE [6, 7] is the penalized version of GEE. This work is inspired by Chen Ling's dissertation [8], and focuses on extending GEE methodology to missing data and informative cluster scenarios through weighting and penalization.

Description

The project develops and implements weighted generalized estimating equation models for correlated data with missing observations. Penalization techniques will be incorporated for high-dimensional covariates, ensuring consistency in estimation and model selection. Performance will be evaluated through extensive simulation studies under different missingness mechanisms and cluster structures. Real datasets will demonstrate the robustness and applicability of WGEE in practical settings. The aim is to provide a unified and computationally feasible framework for analyzing correlated data in the presence of missingness and informative cluster sizes.

Deliverables

- A comprehensive literature review summarizing existing GEE, WGEE, and penalized GEE methodologies, including handling missing data and informative cluster sizes.
- Simulation study results evaluating the performance of WGEE and PWGEE under various missingness mechanisms and correlation structures, with visualizations and summaries.
- Reproducible R scripts and workflows for WGEE and PWGEE.

Prerequisites and skills:

Strong foundation in regression and statistical programming in R is required. nterest in high-dimensional data techniques and penalization methods is advantageous.

References

[1] Ziegler A. Generalized estimating equations. vol. 204. Springer Science and Business Media; 2011.

[2] Wang Y, Paik MC. A weighting approach for GEE analysis with missing data. Statistics in Medicine. 2011;30(16):1925-37.

[3] Salazar-Couso F, et al. Simple GEE and weighted GEE in longitudinal studies with dropouts: Guidelines and implementation in R. BMC Medical Research Methodology. 2016;16:56.

[4] Preisser JS, Lohman K, Rathouz PJ. Performance of weighted estimating equations for longitudinal binary data with dropouts missing at random. Statistics in Medicine. 2002;21(13):2033-46.

[5] Ma Y, Wang H, Jiang X. Penalized weighted GEEs for high-dimensional longitudinal data with informative cluster size. Journal of Multivariate Analysis. 2025. In press.

[6] Niu Y, Wang X, Cao H, Peng Y. Variable selection via penalized generalized estimating equations for a marginal survival model. Statistical Methods in Medical Research. 2020;29(9):2493-506.

[7] Wang L, Zhou J, Qu A. Penalized generalized estimating equations for high-dimensional longitudinal data analysis. Biometrics. 2012;68(2):353-60.

[8] Chen L. Standard and Penalized Generalized Estimating Equations for Longitudinal Data: Methods, Simulations, and Applications. University of Manchester, Department of Mathematics, Faculty of Science and Engineering; 2025. Master's dissertation.

M27: Advanced Missing Data Techniques for Real-World Data Science Applications: A Comprehensive Comparative Evaluation

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Background

Missing data is a common challenge in modern data science, often leading to biased statistical inference, reduced predictive accuracy, and weakened decision-making if not properly addressed. Foundational work by Little and Rubin [1] establishes the core missingness mechanisms-Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR)-and the theoretical consequences of each for valid inference. Van Buuren [2] extends these principles through flexible imputation techniques such as Multivariate Imputation by Chained Equations (MICE), which are particularly suitable for complex, real-world datasets. In more sensitive applications like clinical research, Molenberghs and Kenward [3] emphasize the need for robust modeling strategies and sensitivity analyses when missingness may not be ignorable. This dissertation equips students with a comprehensive understanding of missingness mechanisms, deletion methods, classical imputation approaches, and advanced strategies such as multiple imputation and maximum likelihood. Students will apply these techniques to real datasets, assess the impact of missingness on statistical conclusions, and learn how proper handling preserves validity and statistical power across diverse applied settings.

Description

This project focuses on the development, implementation, and evaluation of multiple imputation strategies for datasets with missing values. Students will explore both traditional statistical methods and modern machine-learning-based approaches. The work includes implementing classical methods such as MICE, predictive mean matching, and maximum likelihood estimation, alongside machine learning techniques including random forest imputation, k-nearest neighbour methods and recent modern approaches. [4, 5] Simulated missingness patterns (MCAR, MAR, MNAR) will be generated to evaluate how different methods perform under controlled scenarios. Performance will be assessed through error metrics, coverage probabilities, and sensitivity analyses. Students will apply the best-performing imputation approach to a real-world dataset of their choice and produce a fully reproducible workflow [4, 6]. This topic has strong potential for extension into Bayesian or deep-learning-based imputation models.

Deliverables

- Literature review summarizing missing-data mechanisms, classical methods, and advanced imputation strategies.
- Implementation of multiple imputation techniques in R or Python, including classical and machinelearning approaches.
- Simulation study evaluating each method under MCAR, MAR, and MNAR conditions.

- Comparative analysis report with performance metrics such as RMSE, accuracy, and confidence interval coverage.
- Sensitivity analyses assessing robustness under different missingness assumptions.
- Fully reproducible scripts and workflows applicable to real-world datasets.

Prerequisites and skills:

Statistical inference, regression analysis, and programming in R or Python.

Optional: Bayesian modeling, machine learning, and experience working with real-world datasets.

References

[1] Little RJA, Rubin DB. Statistical Analysis with Missing Data. 3rd ed. Hoboken, NJ: Wiley; 2019.

[2] van Buuren S. Flexible Imputation of Missing Data. 2nd ed. Boca Raton, FL: CRC Press; 2018.

[3] Molenberghs G, Kenward M. Missing Data in Clinical Studies. Chichester: John Wiley and Sons; 2007.

[4] Chourib I. Missing Data Handling: A Comprehensive Review, Taxonomy, and Comparative Evaluation. Journal of Computer and Communications. 2025;13(6);81-102.

[5] Pereira RC, Abreu PH, Rodrigues PP, Figueiredo MA. Imputation of data Missing Not at Random: Artificial generation and benchmark analysis. Expert Systems with Applications. 2024;249:123654.

[6] Aghdam R, Baghfalaki T, Khosravi P, Saberi Ansari E. The ability of different imputation methods to preserve the significant genes and pathways in cancer. Genomics, proteomics and bioinformatics. 2017;15(6):396-404.

M28: Linking Longitudinal Trajectories to Event Risks: A Joint Modeling Approach

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Background

In many longitudinal studies, repeated measurements are recorded until an event occurs, such as disease progression or death. Analyzing longitudinal biomarkers and time-to-event outcomes separately may lead to biased estimates and loss of efficiency because it ignores the association between the biomarker trajectory and the risk of the event. Joint modeling provides a unified framework to analyze both processes simultaneously, capturing their interdependence and allowing for accurate inference [1 – 3]. Joint models typically link the longitudinal and time-to-event processes through shared random effects or latent variables, which enables estimation of both the evolution of biomarkers and the hazard function for the event. They also facilitate dynamic prediction[4], allowing patient-specific risk predictions to be updated as new longitudinal measurements become available[5, 6]. These models are widely used in clinical and biomedical research, with implementations available in R via packages such as JM, JMbayes, and joineR.

Description

The project aims to develop and apply joint models for longitudinal and time-to-event outcomes, exploring their interdependence and predictive value. Simulation studies will evaluate estimation accuracy, bias, and efficiency under various scenarios, including different correlation strengths, censoring mechanisms, and sample sizes. Real clinical datasets will be analyzed to illustrate the advantages of joint modeling for dynamic prediction and personalized decision-making. Extensions may include high-dimensional covariates, penalization techniques, multilevel/hierarchical structures, zero-inflation, competing risks, and cure-rate data, providing a flexible framework for complex longitudinal-survival studies[7–11, 11–13].

Deliverables

- A comprehensive literature review on joint modeling of longitudinal and time-to-event data.
- Simulation study results evaluating the performance of joint models with visualizations and summaries.
- Reproducible R scripts and workflows for fitting and validating joint models, including dynamic prediction tools for individualized risk assessment..

Prerequisites and skills:

A solid understanding of regression is required. Proficiency in statistical programming (preferably R) is essential. Familiarity with mixed-effects models, or Bayesian methods is advantageous.

References

[1] Rizopoulos D. Joint Models for Longitudinal and Time-to-Event Data: With Applications in R. Chapman Hall/CRC; 2012.

[2] Taylor JMG. Joint modeling of longitudinal and survival data: an overview. Statistics in Medicine. 2013;32(8):1295-311.

[3] Henderson R, Diggle P, Dobson A. Joint modelling of longitudinal measurements and event time data. Biostatistics. 2000;1(4):465-80.

[4] Hashemi R, Baghfalaki T, Philipps V, Jacqmin-Gadda H. Dynamic prediction of an event using multiple longitudinal markers: a model averaging approach. Statistics in Medicine. 2025;44(1314):e70122.

[5] Rizopoulos D. Dynamic predictions and prospective accuracy in joint models for longitudinal and time-to-event data. Biostatistics. 2011;12(3):333-44.

[6] Komsta L. Dynamic predictions in joint models for longitudinal and time-to-event data. Journal of Statistical Software. 2021;98(5):1-25.

[7] Li Y, et al. High-dimensional joint modeling of longitudinal and survival data with penalization. Statistics in Medicine. 2017;36(10):1609-25.

[8] Taylor JMG, et al. Joint models for longitudinal and survival data with competing risks. Statistical Methods in Medical Research. 2014;23(3):295-310.

[9] Baghfalaki T, Ganjali M. Bayesian Joint Modeling of Zero-Inflated Longitudinal Data and Survival with a Cure Fraction. arXiv preprint arXiv:250819001. 2025.

[10] Ariyo O, Olobatuyi K, Baghfalaki T. A Bayesian joint bent-cable model for longitudinal measurements and survival time with heterogeneous random-effects distributions. Journal of Biopharmaceutical Statistics. 2025.

[11] Baghfalaki T, Hashemi R, Tzourio C, Helmer C, Jacqmin-Gadda H. A Two-stage Approach for Variable Selection in Joint Modeling of Multiple Longitudinal Markers and Competing Risk Outcomes. Statistical Methods in Medical Research. 2025.

[12] Ganjali M, Baghfalaki T, Balakrishnan N. Joint modeling of zero-inflated longitudinal measurements and time-to-event outcomes with applications to dynamic prediction. Statistical Methods in Medical Research. 2024.

[13] Baghfalaki T, Ganjali M, Martins R. Approximate Bayesian inference for joint partially linear modeling of longitudinal measurements and spatial time-to-event data. Journal of Statistical Computation and Simulation. 2025;95(16):3548-76.

M29: Bayesian Models for Detecting Pleiotropy Across Multiple Traits

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Background

Pleiotropy, in which a single genetic variant influences multiple phenotypes, is fundamental to understanding shared biological mechanisms across complex diseases. In genomics, identifying variants, genes, or pathways that simultaneously affect multiple traits can reveal common mechanisms, enhance risk prediction, and guide therapeutic development. Traditional GWAS analyses often consider each trait independently, overlooking correlations between traits and thereby reducing power to detect shared genetic effects. Bayesian hierarchical models provide a principled approach for integrating high-dimensional summary statistics across multiple traits while capturing structured genetic relationships, such as gene sets or pathway membership. A key motivating contribution is the GCPBayes model of [1], which uses group-structured priors to detect pleiotropic signals across cancers using group-level genomic information. Recent methodological advances include the group-structured pleiotropy meta-analysis of [2], sparse multi-trait Bayesian models [3], multi-trait GWAS meta-analysis (MTAG) [4], and empirical-Bayes local false discovery rate approaches [5], highlighting the growing interest in leveraging shared genetic architecture for cross-trait analysis.

Description

This project develops and applies Bayesian models for detecting pleiotropic genetic effects across multiple traits. The framework builds on hierarchical priors to identify effects at both the variant and pathway levels. Potential methodological extensions include:

- Novel structured priors for gene-set, pathway-level, or functional annotation pleiotropy.
- Decomposition of genetic effects into trait-specific and shared components.
- Scalable Bayesian inference using variational methods or pseudo-marginal MCMC.
- Application to multi-trait GWAS summary datasets for biologically interpretable pleiotropic signal detection.

Simulation studies will evaluate estimation accuracy, detection power, and robustness under varying effect-size heterogeneity, correlation structure, and sample sizes. Real-world data analyses will illustrate the model's utility in uncovering shared genetic architecture across traits.

Deliverables

- A comprehensive literature review summarizing existing Bayesian methods for multi-trait GWAS and pleiotropy detection, including structured priors and hierarchical modeling approaches.
- Simulation study results assessing the performance of the proposed Bayesian models under various genetic architectures, effect-size heterogeneity, and correlation structures, with visualizations and summaries.

- Reproducible R and/or Python scripts implementing the Bayesian pleiotropy models, including preprocessing of multi-trait GWAS summary statistics, model fitting, and visualization of pleiotropic effects at both variant and pathway levels.

Prerequisites and skills:

A strong foundation in regression and Bayesian modelling is required. Proficiency in statistical programming (R or Python) is essential. Interest in high-dimensional genomics and hierarchical Bayesian computation is advantageous.

References

[1] Baghfalaki T, Sugier PE, Truong T, Pettitt AN, Mengersen K, Liquet B. Bayesian meta-analysis models for cross-cancer genomic investigation of pleiotropic effects using group structure. Statistics in Medicine. 2021;40(6):1498-518.

[2] Sugier PE, Asgari Y, Sedki M, Truong T, Liquet B. Meta-analysis models with group structure for pleiotropy detection at gene and variant level using summary statistics from multiple datasets. Biostatistics. 2025;26(1):kxat037.

[3] Maity AK, Hellinger N, Xu Z. Bayesian sparse multi-phenotype association modelling for pleiotropic genetic effects. Genetic Epidemiology. 2022.

[4] Turley P, Walters RK, et al. Multi-trait analysis of genome-wide association summary statistics using MTAG. Nature Genetics. 2018;50:229-37.

[5] Golan D, Lander ES, Rosset S. Measuring pleiotropy via the local false discovery rate. PLoS Genetics. 2014;10(3):e1004722.

M30: Integrative Statistical Modeling of Multi-Omics Data

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Background

Omics data refers to comprehensive, high-dimensional molecular measurements that capture the various layers of biological systems. This includes genomics (DNA-level data), transcriptomics (gene expression), proteomics (proteins), metabolomics (metabolites), and epigenomics (DNA modifications and regulatory signals). Such data provide a system-wide view of cellular processes, enabling the study of disease mechanisms, biomarker discovery, and patient stratification [1–3]. Multi-omics integration involves combining multiple omics layers to reveal relationships and interactions that may not be visible from a single layer. This presents statistical challenges due to high dimensionality, heterogeneity, and correlation between layers. In certain clinical contexts, omics measurements are collected longitudinally, and joint modeling frameworks can link these biomarkers with patient outcomes to improve prediction and inference [4].

Description

The project will develop statistical methods for integrating and analyzing multi-omics data. Key objectives include:

- Preprocessing and normalization of multi-omics datasets from publicly available repositories.
- Integration across omics layers using penalized regression, dimension reduction, and factor analysis methods.
- Identification of molecular signatures and pathway-level patterns predictive of phenotypes or clinical outcomes.
- Evaluation via simulation studies under varying noise, missingness, and correlation structures.
- Optional exploration of dynamic prediction using joint modeling of longitudinal biomarkers and survival outcomes.

This approach will provide insights into cross-omics interactions and support translational applications in disease research.

Deliverables

- A comprehensive literature review on multi-omics integration methods.
- Simulation study results evaluating the performance of the proposed statistical methods under varying levels of noise, missing data, and cross-omics correlation structures.
- Reproducible R and/or Python scripts for preprocessing, integrating, and analyzing multi-omics datasets, including identification of predictive molecular signatures and pathway-level patterns.

- Application of the developed methods to at least one real multi-omics dataset, demonstrating the utility of the approach for phenotype prediction or clinical outcome association.

Prerequisites and skills:

Knowledge of regression, multivariate statistics, and high-dimensional modeling. Proficiency in statistical programming (R, or Python). Experience with omics data preprocessing and integration is advantageous.

References

[1] Goh WWB, Wong L. A comprehensive survey of computational methods for multi-omics data integration. Briefings in Bioinformatics. 2020;21(2):407-25.

[2] Huang S, Chaudhary K, Garmire LX. More is better: Recent progress in multi-omics data integration methods. Frontiers in Genetics. 2021;12:1-15.

[3] Argelaguet Rea. Multi-Omics Factor Analysis-a framework for unsupervised integration of multiomics data sets. Molecular Systems Biology. 2018;14(6):e8124.

[4] Rizopoulos D. Joint Models for Longitudinal and Time-to-Event Data: With Applications in R. Chapman Hall/CRC; 2012.

M31: Online Controlled Experiments

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Background

Major websites such as Amazon, Facebook, Google, Netflix, and so on run online controlled experiments on millions of users every day to test and optimize various aspects of their operations. For example, several years ago Google ran a large experiment to test the best shade of blue for links, resulting in a $200 million dollar increase in revenue, though most changes typically result in much smaller gains. These experiments, also known as âĂIJA/B testsâĂİ, are cheap to run and generate enormous volumes of data. However, they still require careful design and statistical analysis to be effective.

Description

In this project, students will explore some of the challenges in online controlled experiments and learn about the statistical methodology available to deal with these challenges (see e.g. [1]). The application of the techniques can be explored via open data sets, e.g. [2].

One potential topic is optional stopping: it can be desirable to terminate the experiment early if it becomes clear that the new variant of the website has a very large effect (either positive or negative). However, optional stopping requires careful implementation to avoid inflating the Type I error rate. Methods include the mixture sequential probability ratio test, as implemented in Optimizely [3], Bayesian methods, as used by AutoTrader, and group sequential tests [4], as used by Spotify. Students could explore these methods in detail and their application to an example data set.

Deliverables

- Hand-coded implementations of sequential testing methods
- Simulation studies exploring the statistical properties of the different methods (Type I error rate, power etc.)
- Illustration of methods via application to a real data set

Prerequisites and skills:

Good coding skills in R or Python. An understanding of statistical inference concepts such as likelihood ratio tests, Type I error rates, power, and Bayesian methods, e.g. as discussed in Statistics and Machine Learning 1.

References

[1] Larsen, N. et al. (2023) âĂŸStatistical Challenges in Online Controlled Experiments: A Review of A/B Testing MethodologyâĂŹ, The American Statistician, 78, 135âĂŞ149. https://dx.doi.org/10.1080/00031305.2023.2257237

[2] Liu, C.H.B. et al. (2021) âĂŸDatasets for Online Controlled ExperimentsâĂŹ, 35th Conference on Neural Information Processing Systems (NeurIPS 2021) Track on Datasets and Benchmarks, https://arxiv.org/abs/2111.10198

[3] Johari, R. et al. (2017) âĂŸPeeking at A/B Tests: Why it matters, and what to do about itâĂŹ In KDDâĂŹ17: Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, New York, 1517âĂŞ1525. https://doi.org/10.1145/3097983.3097992

[4] Kim, K. and Tsiatis, A.A. âĂIJIndependent increments in group sequential tests: a reviewâĂİ. SORT-Statistics and Operations Research Transactions, 44, 223-64, https://dx.doi.org/ 10.2436/20.8080.02.101

M32: Bayesian experimental design

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Background

A critical but often under-appreciated question in Data Science is "what data should we collect?". This is especially important in experimental contexts where often only a limited number of runs can be performed. One way to solve this problem is to construct an optimal Bayesian experimental design.

Specifically, suppose that the experiment aims to investigate the relationship between controllable variables, x, and a response, y, by observing responses $y_1, \ldots, y_n$ at a collection of design points $x_1, \ldots, x_n$. Suppose also that we have available a model $p(y|x, \theta)$ that describes the distribution of y given x and parameter values θ, and a prior distribution with density $p(\theta)$ describing our beliefs about the parameters. Then one way to choose a Bayesian optimal design is to find a $\xi = (x_1, \ldots, x_n)$ that maximizes the expected information gain utility,

$$U(\xi) = E\left[\log \frac{p(\theta|x_{1:n}, y_{1:n})}{p(\theta)}\right],$$

which can be thought of as an expected "distance" between the prior and posterior distributions. However, actually finding the design that maximizes this utility is quite challenging and requires several advanced methods, such as stochastic gradient optimization, variational approximations, nested or multilevel Monte Carlo, and reinforcement learning.

Description

In this project, which is more methodological, the student will work to understand modern techniques for the construction of Bayesian optimal designs, implement some of the techniques computationally, and find new designs for some problems in the literature. It could provide a springboard for further research in this cutting-edge topic.

Deliverables

- A summary of modern algorithms for Bayesian experimental design
- Programming implementations of techniques for approximating expected utility and optimizing designs
- Illustration of methods via application to real-world examples from the literature, e.g. the crystallography experiment in [2]

Prerequisites and skills:

Strong programming skills in Python. A good understanding of Monte Carlo techniques and Deep Learning ideas as discussed in MATH64071 Introduction to Uncertainty Quantification and DATA70132 Statistics and Machine Learning 2. Experience with Pytorch and/or Pyro would be advantageous.

References

[1] Rainforth, T. et al (2024) Modern Bayesian experimental design, *Statistical Science*, 39, 100-114. https://dx.doi.org/10.1214/23-STS915

[2] Woods, D.C. et al (2006) Designs for Generalized Linear Models With Several Variables and Model Uncertainty, *Technometrics*, 48, 284âĂŞ292. https://doi.org/10.1198/004017005000000571

M33: Managing storage in renewable energy networks

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Background

A key challenge for the incorporation of renewable energy into energy networks is how to account for the variability in production while maintaining the reliability that is needed by users. This requires storing energy and the amount that should be stored at any time depends on the future demand (largely predictable but varying over the course of a day) and future production (highly uncertain and depending on weather). The goal of maintaining system reliability while minimising energy loss becomes an optimisation problem that is complicated by future uncertainty.

Description

The purpose of this project is to explore strategies for operating storage facilities from the perspective of an overseer who wants to ensure there is enough supply to meet demand over time while minimising wasted energy.

Deliverables

- Finding and interpreting data from the National Grid on renewable energy production and consumer demand.
- Producing code that can take in this data and test different energy storage strategies.
- Analysis based on the output of this code.

Prerequisites and skills:

The project will require mathematical and programming skills.

References

[1] A. I. Bejan, R. J. Gibbens and F.P. Kelly. Statistical Aspects of Storage Systems Modelling In Energy Networks. 46th Annual Conference on Information Sciences and Systems (CISS), 2012.

M34: Kalman filter with application to finanical state-space models

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Background

State space model deals with a set of states changing over time, such as the relationship between pairs of cointegrated equities. However, observations of these states are often distorted by statistical noise, preventing direct access to their true values. The aim of the model is to deduce information about these states based on the available, albeit noisy, observations, and to update this understanding as new data comes in.

State space models are highly versatile and can encompass various time series models, including ARMA, ARIMA, and GARCH. A key algorithm in this context is the Kalman Filter, a renowned algorithm with applications in both engineering and quantitative finance. In engineering, it's used for control problems like navigation and spacecraft trajectory analysis, while in finance, it aids in analyzing financial time series.

The model focuses on three main types of inference: prediction (forecasting future state values), filtering (estimating current state values from past and current observations), and smoothing (estimating past state values from all available observations). The Kalman Filter is particularly effective in performing these inferential tasks.

Description

In this project you will learn the concept of the state-space model (SSM), the architecture of formulating SSM, and the inferential concepts of filtering, smoothing and prediction. You will learn the Kalman filter to carry out these tasks. You will apply the Kalman filter to estimate time-varying betas in the context of estimating the capital asset pricing model(CAPM).

Deliverables

- Review of the literature on state-space models and its filtering, smoothing and prediction.
- To learn and implement Kalman filter to carry out these tasks.
- Extracting and processing financial time series from public domains.
- Estimating CAPM with time-varying betas.

Prerequisites and skills:

- Experience and willingness to carry out intensive programming using R or Python.

References

[1] Durbin, James, and Siem Koopman (2012) *Time series analysis by state space methods.*, OUP Oxford.

[2] Brooks, Chris (2019) *Introductory econometrics for finance*, Cambridge university press.

[3] `www.quantstart.com/articles/Dynamic-Hedge-Ratio-Between-ETF-Pairs-Using-the-Kalman`

M35: Estimating District-Level Minority Voting Behaviour

Proposer: Thomas House and Jack Bailey
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Background

After every general election, the British Election Study runs a "gold standard" face-to-face random probability survey to determine how the British electorate voted. However, this survey typically samples only around 3,000 people. Since we are often interested in how subgroups behave, this is a serious problem: we lack the statistical power to say much at all about how minorities have voted and how their behaviour may have differed from district to district. This also has major political implications, since it prevents parties from being able to understand or effectively target their supporters and means that concerns facing minorities may not be picked up in the media.

Description

The purpose of this project is to try to overcome this problem by computing constituency-level estimates of minority voting behaviour using *multilevel regression and post-stratification* (or some other well-motivated approach, for instance, using model averaging or machine learning). The student can choose which minority group they are most interested in, but the most politically-relevant cases right now are probably either ethnic or sexual minorities.

Deliverables

- Understand multilevel regression and post-stratification, its advantages and its shortcomings, and apply it to data from the British Election Study Internet Panel
- Produce a reproducible pipeline that outputs a data set of constituency-level estimates of party support for your chosen minority group at the 2024 UK general election in Great Britain (i.e. excluding Northern Ireland)

Prerequisites and skills:

A passing understanding of British politics is essential, as is an understanding of multilevel regression and / or model averaging & machine learning, and the ability to implement these models in some programming language such as R, Python, or Julia. A handful of useful articles are suggested below.

References

[1] Ford, R., Bale, T., Jenning, W., and Surridge, P. (2025). *The British General Election of 2024*. London: Palgrave Macmillan.

[2] Hanretty, C. (2020). An Introduction to Multilevel Regression and Post-Stratification for Estimating Constituency Opinion. *Political Studies Review*, 18 (4), pp. 630-645.

[3] Hanretty, C., Lauderdale, B.E., and Vivyan, N. (2018). Comparing Strategies for Estimating Constituency Opinion from National Survey Samples. *Political Science Research and Methods*, 6 (3), pp. 571-591.

[4] Ornstein, J. (2020). Stacked Regression and Post-Stratificatino. *Political Analysis*, 28 (2), pp. 293-301.

M36: Bayesian Gaussian Process Regularisation

Proposer: Thomas House
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Background

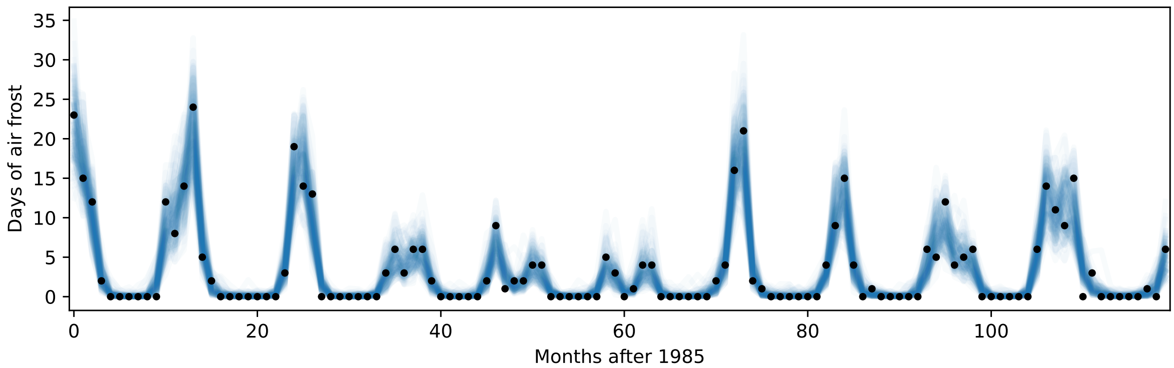
Gaussian processes (GPs) are a powerful modelling tool in data science [**GPbook**]. In the most straightforward case they can be motivated in a frequentist or Bayesian framework, but for the case of more general response,

Description

Suppose we have a set of feature vectors $\{\mathbf{x}_i\}$ and responses $\{y_i\}$. These responses might be continuous, continuous and positive, integer, or 0-1. For the integer case, Bayesian Gaussian process regularisation of the problem might involve modelling $y_i \sim \mathrm{Poisson}(\exp(z_i))$ and asserting a GP prior on the latent variables $\mathbf{z} = (z_i)$ with pdf

$$\pi(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathcal{K}(\{\mathbf{x}_i\}))\,.$$

In this case, Bayesian methods can be used to treat the elements of $\mathbf{z}$, which are not directly observed, as parameters, in a procedure related to multiple imputation. This project will consider implementation of such a process with applications in COVID variant surveillance [**ONSlin**]. An example of taking such an approach to open meteorological data is shown below.



Deliverables

- Derivation of relevant priors and likelihoods for survival, Bernoulli and other models.
- Implementation of the model in a probabilistic programming language such Stan.
- Application to real data, particularly on COVID variants.

Prerequisites and skills

This should be accessible to any student on the mathematics pathway, but either experience of or willingness to learn probabilistic programming is essential.

References

[1] Carl Edward Rasmussen and Christopher K. I. Williams (2006) *Gaussian Processes for Machine Learning*, MIT Press gaussianprocess.org

[2] Katrina A. Lythgoe, Tanya Golubchik, Matthew Hall, Thomas House, . . . (2023) "Lineage replacement and evolution captured by 3 years of the United Kingdom Coronavirus (COVID-19) Infection Survey." *Proceedings of the Royal Society B* 10.1098/rspb.2023.1284.

M37: Inferring attitude change from internet posts

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Background

Spotting potentially dangerous radicalization of individuals and prioritising resources to manage threats is an important part of the UK National Security operation. QinetiQ has data on internet posts that have each been classified into one of three categories: central, extreme or potentially dangerous. Similar data is available on posts regarding Brexit, and this mgiht also provide the focus for the research. This project will use data from QinetiQ to examine whether simple modelling of internet posts using this classification can help understand the evolution of attitudes and behaviour.

This is part of a project with colleagues at the University of Warwick and Exeter, and at QinetiQ.

Description

The purpose of this project is to explore modelling strategies to understand data relating to the characterisation of internet posts. In particular whether it is possible to identify patterns of posts that can pressage the extreme radicalization of individuals on a partricular issue. (It is important to note that the expression of extreme views on their own are not considered a reason for concern.)

Such modelling could help the UK National Security effort, but there are clearly ethical issues around the data and the modelling that also need to be addressed in the project.

Deliverables

- A brief introduction to the data available and the simple modelling strategies that will be adopted (e.g. Markov compartmental models).
- An initial (simple) approach to identifying stationary distributions and transition rates for the Markov models from the data.
- Is there evidence that these rates change as a function of political events? Can this be done at the population level of the data and/or for individuals?
- Development of more sophisticated models using the data (e.g. refined compartments, memory).
- Exploration of ethical issues raised by the project.

Prerequisites and skills

A basic understanding of stochastic modelling, particularly Markov processes and/or epidemic modelling. Reasonable programming skills.

References

[1] Froyland, G., Judd, K., Mees, A.I., Watson, D. & Murao, K. (1995) *Constructing invariant measures from data*, International Journal of Bifurcation and Chaos, 5, 1181-1192

[2] Gibson, G.J. & Renshaw, E. (1998) *Estimating parameters in stochastic compartmental models using Markov chain methods*, IMA Journal of Mathematics Applied in Medicine & Biology, 15, 19-40, https://doi.org/10.1093/imammb/15.1.19.

[3] Lane, R.O., Holmes, W.J., Taylor, C.J., State-Davey, H.M. & Wragge, A.J. (2021) *Predicting the descent into extremism and terrorism*, Mathematics in Defence and Security, Online Conference, 30-31 March 2021. https://arxiv.org/abs/2509.16014.

[4] Lane, R.O., State-Davey, H.M., Taylor, C.J., Holmes, W.J., Boon, R.A. & Round, M.D. (2023) *Behavioural Analytics: Mathematics of the Mind,* Mathematics in Defence and Security, London, UK, 7th September 2023. https://arxiv.org/abs/2502.00013.

[5] Meylahn, B.V., De Turck, K., & Mandjes, M. (2024) *Trust in society: A stochastic compartmental model*, https://arxiv.org/pdf/2409.12686v1.

M38: Asynchronous updating for differential equations

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Background

It can be expensive (in computer time and memory) to solve high dimensional ordinary differential equations numerically. One way to cut down on the number of computations required is to update variables asynchronously, so at each timestep only a subset of the full system is updated whilst the remaining variables are treated as constant. This is a standard technique, called the Gillespie algorithm, in the study of chemical reactions but it is less well developed for general systems of differential equations.

This is part of a project with colleagues at the University of Bath and Jonas Latz here in Manchester.

Description

The purpose of this project is to explore possible asynchronous models for solving differential equations and to assess the efficiency and accuracy of these methods. The main example used will be the Lorenz equations and the partial differential equations from which they are derived. The Lorenz equations were one of the earliest examples of chaotic motion and their original derivation was via a Galerkin truncation of a model of double diffusive convection in the atmosphere. As the number of Galerkin modes used in the simulation increases, the chaotic motion is pushed to higher Rayleigh number and there seems to be no chaos in the full partial differential equations. This change in behaviour, and the understanding we have of the low dimensional model, makes it a good candidate for the study of possible asynchronous simulations.

There is also an analogy with recent work by Jonas Latz. Latz considers randomizing the timestep of the simulation and proves results for the Euler method under this modification. It is likely that some of the ideas from this analysis will help understand the asynchronous updating model.

Deliverables

- A brief survey of the numerical techniques and existing models of asynchronous updating in large scale simulations.
- An introduction to the Lorenz equations and their derivation. Consideration of the possible lower order truncations and their energy. Numerical verification of the threshold of chaos as the number of Galerkin modes is increased.
- Numerical experiments (and theory if possible) on low dimensional asynchronous updating. Are there results about approach to a neighbourhood of the deterministic attractor?
- Numerical experiments (and theory if possible) on high dimensional asynchronous updating. How should probabilities be chosen? How does this relate to the low dimensional models?

Prerequisites and skills

A basic understanding of the analysis of differential equations and how they are simulated. Some experience of stochastic modelling would be useful. Reasonable programming skills.

References

[1] Saltzman, B. (1962) *Finite Amplitude Free Convection as an Initial Value Problem-I*, J. Atmos. Sci., 19, 329-341, https://doi.org/10.1175/1520-0469(1962)019<0329:FAFCAA>2.0.CO;2.

[2] Lorenz, E. N. (1963) *Deterministic Nonperiodic Flow*, J. Atmos. Sci., 20, 130-141, https://doi.org/10.1175/1520-0469(1963)020<0130:DNF>2.0.CO;2.

[3] Marcus, P.S. (1981) *Effects of truncation in modal representations of thermal convection*, Journal of Fluid Mechanics. 103, 241-255. doi:10.1017/S0022112081001328.

[4] Gillespie, D.T. (2007) *Stochastic Simulation of Chemical Kinetics*, Annual Review of Physical Chemistry, 58, 35-55. doi:10.1146/annurev.physchem.58.032806.104637.

[5] Latz, J. (2025) *The random timestep Euler method and its continuous dynamics*, IMA Journal of Numerical Analysis, https://doi.org/10.1093/imanum/draf096.